

Dual Integral Quadratic Constraints for Robust Control of Partial Integral Equations

Thijs Lenssen

Abstract—

Index Terms—PDEs, Robust Control, LMIs;

I. INTRODUCTION

The broader system-theoretic foundations include the semi-group treatments of Curtain and Pritchard [3] and Curtain and Zwart [4], and the well-posed linear systems framework of Staffans [28]. Earlier infinite-dimensional robust control includes stabilization by finite-dimensional controllers [2] and state-space H_∞ synthesis for distributed-parameter systems [33].

Related PDE methods include backstepping [9], robust Lyapunov-based control [6], and semidefinite stability analysis [32]. PIE representations provide an operator framework for PDE analysis [15], [25], with duality enabling nominal state-feedback synthesis [26].

IQCs describe uncertainty [11] and connect to storage inequalities through dissipativity [23], [38]. PIE-based developments cover robust ODE–PDE analysis [5], infinite-dimensional hard IQCs [29], and structured robustness, performance, and observer synthesis [10].

Dual IQCs support robust feedforward design [8] and LPV estimator and feedforward synthesis [36]. Convex synthesis depends on the system structure [20]; general dynamic multiplier methods use alternating design and analysis [35], [37]. Dynamic-IQC feedback also addresses uncertain systems with state delay [39].

For finite-horizon certificates, relevant connections include doubly-hard factorizations [1], IQCs with terminal costs [22], and multiplier storage that reduces conservatism [7], [18].

[30]

II. NOTATION & PRELIMINARIES

A. Notation

$L_2^n[a, b]$ denotes the set of $\mathbb{R}^n$ -valued, Lebesgue square-integrable equivalence classes of functions on the spatial domain $[a, b] \subset \mathbb{R}$. The space $\mathbb{R}L_2^n[a, b]$ denotes the Cartesian product $\mathbb{R}^{n_1} \times L_2^{n_2}[a, b]$, where $\mathbf{n} = \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$, with inner product

$$\left\langle \begin{bmatrix} x_1 \\ \mathbf{x}_2 \end{bmatrix}, \begin{bmatrix} y_1 \\ \mathbf{y}_2 \end{bmatrix} \right\rangle_{\mathbb{R}L_2} := x_1^\top y_1 + \langle \mathbf{x}_2, \mathbf{y}_2 \rangle_{L_2}.$$

T. Lenssen and A. Das are with the Control Systems Group, Eindhoven University of Technology, The Netherlands. Email addresses: t.m.lenssen@student.tue.nl, am.das@tue.nl

Corresponding author: am.das@tue.nl

When clear from context, we occasionally omit the subscript and simply write $\langle \cdot, \cdot \rangle := \langle \cdot, \cdot \rangle_{\mathbb{R}L_2}$. Since the only spatial domain considered here is $[a, b]$, we typically omit the domain and write L_2^n or $\mathbb{R}L_2^n$, further omitting dimensions when clear from context. For a normed space X and Banach space Y , $\mathcal{L}(X, Y)$ denotes the Banach space of bounded linear operators from X to Y with induced norm

$$\|\mathcal{P}\|_{\mathcal{L}(X, Y)} = \sup_{\|\mathbf{x}\|=1} \|\mathcal{P}\mathbf{x}\|_Y,$$

and $\mathcal{L}(X) := \mathcal{L}(X, X)$. Our notational convention is to write functions in bold (e.g., $\mathbf{x}$, indicating $\mathbf{x} \in \mathbb{R}L_2$) and operators in calligraphic font (e.g., $\mathcal{P} \in \mathcal{L}(\mathbb{R}L_2)$). For a Hilbert space X and operator $\mathcal{A} \in \mathcal{L}(X)$, the adjoint $\mathcal{A}^*$ is defined by

$$\langle \mathbf{x}, \mathcal{A}\mathbf{y} \rangle_X = \langle \mathcal{A}^*\mathbf{x}, \mathbf{y} \rangle_X, \quad \forall \mathbf{x}, \mathbf{y} \in X.$$

The space $L_2^n[0, \infty)$ denotes $\mathbb{R}^n$ -valued square-integrable signals defined on the temporal domain $[0, \infty)$, so that $x(t) \in \mathbb{R}^n$. Similarly, $\mathbb{R}L_2^n[0, \infty)$ denotes $\mathbb{R}L_2$ -valued square-integrable temporal signals. For suitably differentiable $\mathbf{x} \in \mathbb{R}L_2^n[0, \infty)$, we write $\dot{\mathbf{x}} := \frac{\partial \mathbf{x}}{\partial t}$. For a Hilbert space $\mathbf{H}$, we denote by $\mathbf{L}_{\mathbf{H}}$ the space of square-integrable functions $u : [0, \infty) \rightarrow \mathbf{H}$ with inner product

$$\langle \mathbf{u}, \mathbf{v} \rangle_{\mathbf{L}_{\mathbf{H}}} = \int_0^\infty \langle \mathbf{u}(t), \mathbf{v}(t) \rangle_{\mathbf{H}} dt.$$

We define the extended space $\mathbf{L}_{e, \mathbf{H}}$ to consist of functions $u : [0, \infty) \rightarrow \mathbf{H}$ such that

$$\int_0^T \|\mathbf{u}(t)\|_{\mathbf{H}}^2 dt < \infty \quad \forall T \geq 0.$$

We use $\mathbf{L}_{e, [a, b]}$ and $\mathbf{L}_{[a, b]}$ to denote $\mathbf{L}_{e, \mathbf{H}}$ and $\mathbf{L}_{\mathbf{H}}$ with $\mathbf{H} = \mathbb{R}L_2[a, b]$. For any Hilbert space $\mathbf{H}$, the truncation operator $p_T : \mathbf{L}_{e, \mathbf{H}} \rightarrow \mathbf{L}_{e, \mathbf{H}}$ is defined for $\mathbf{y} \in \mathbf{L}_{e, \mathbf{H}}$ by

$$(p_T \mathbf{y})(t) = \begin{cases} \mathbf{y}(t), & 0 \leq t \leq T, \\ 0, & \text{otherwise.} \end{cases}$$

such that $(p_T \mathbf{y})(t) \in \mathbf{L}_{\mathbf{H}}$ for all $T \geq 0$.

Definition 1. Let $\mathbf{H}$ and $\mathbf{G}$ be Hilbert spaces. For all $T \geq 0$, an operator $G : \mathbf{L}_{\mathbf{H}} \rightarrow \mathbf{L}_{\mathbf{G}}$ is:

- Causal if $p_T G = p_T G p_T$.
- Bounded on $\mathbf{L}_{\mathbf{H}}$ if $\|G\mathbf{v}\|_{\mathbf{L}_{\mathbf{G}}} \leq C\|\mathbf{v}\|_{\mathbf{L}_{\mathbf{H}}}$ for some $C \geq 0$ and all $\mathbf{v} \in \mathbf{L}_{\mathbf{H}}$.
- Bounded on $\mathbf{L}_{e, \mathbf{H}}$ if $\|p_T G\mathbf{v}\|_{\mathbf{L}_{\mathbf{G}}} \leq C\|p_T \mathbf{v}\|_{\mathbf{L}_{\mathbf{H}}}$ for some $C \geq 0$ and all $\mathbf{v} \in \mathbf{L}_{e, \mathbf{H}}$.

Finally, we will note some operations. For a given $\mathcal{K} \in \mathcal{L}(\mathbb{R}L_2^n, \mathbb{R}L_2^m)$, we define the operator $\mathcal{M}_{\mathcal{K}} : \mathbf{L}_{e, [a, b]}^n \rightarrow$

$\mathbf{L}_{e,[a,b]}^m$ to compactly represent a multiplication operator. For $\mathbf{u} = [u_1 \cdots u_N]^\top$ with $\mathbf{u} \in \mathbb{R}L_2$, its action at time t is given by the block structure of $\mathcal{K}_{ij}$:

$$(\mathcal{M}\mathbf{u})(t) := \begin{bmatrix} \mathcal{K}_{11} & \cdots & \mathcal{K}_{1n} \\ \vdots & \ddots & \vdots \\ \mathcal{K}_{m1} & \cdots & \mathcal{K}_{mn} \end{bmatrix} \begin{bmatrix} \mathbf{u}_1 \\ \vdots \\ \mathbf{u}_n \end{bmatrix} (t).$$

B. Partial Integral Equations (PIEs)

Partial Integral operators, also known as PI operators, are a class of bounded linear integral operators that are defined jointly on a vector space and Hilbert space [15], [25].

Definition 2. We say $\mathcal{P} \in \Pi_4^{m \times n} \subset \mathcal{L}(\mathbb{R}L_2^n, \mathbb{R}L_2^m)$, denoted by $\Pi \left[\begin{array}{c|c} P & Q_1 \\ \hline Q_2 & \{R_0, R_1, R_2\} \end{array} \right]$, if there exist a matrix P and matrix polynomials Q_1, Q_2, R_0, R_1 , and R_2 (of compatible dimension) such that,

$$\begin{aligned} & \left(\Pi \left[\begin{array}{c|c} P & Q_1 \\ \hline Q_2 & \{R_0, R_1, R_2\} \end{array} \right] \right) \begin{bmatrix} x \\ \mathbf{x} \end{bmatrix} (s) \\ & := \begin{bmatrix} Px + \int_a^b Q_1(s) \mathbf{x}(s) ds \\ Q_2(s)x + \mathcal{R}\mathbf{x}(s) \end{bmatrix}, \\ & (\mathcal{R}\mathbf{x})(s) = R_0(s)\mathbf{x}(s) + \int_a^s R_1(s, \theta)\mathbf{x}(\theta) d\theta \\ & \quad + \int_s^b R_2(s, \theta)\mathbf{x}(\theta) d\theta. \end{aligned}$$

The vector space of PI operators, given compatible dimensions, are closed under composition, addition, adjoint and concatenation – implying that Π_4 is a composition algebra [25].

Definition 3. (LPIs) For given PI operators $\{\mathcal{E}_{i,j}, \mathcal{F}_{i,j}, \mathcal{G}_i\}$ and a convex linear functional $\mathcal{L}(\cdot)$, a linear PI inequality is a convex optimization of the following form

$$\begin{aligned} & \min_{P_i, Q_{1i}, Q_{2i}, R_{0i}, R_{1i}, R_{2i}} \mathcal{L}(\{P_i, Q_{1i}, Q_{2i}, R_{0i}, R_{1i}, R_{2i}\}) \\ & \text{s.t.} \sum_{j=1}^K \mathcal{E}_{ij}^* \Pi \left[\begin{array}{c|c} P_i & Q_{1i} \\ \hline Q_{2i} & \{R_i\} \end{array} \right] \mathcal{F}_{ij} + \mathcal{G}_i \succeq 0 \end{aligned}$$

Only LMIs are required to solve an optimization problem involving LPIs. These definitions are based on [26]

Definition 4 (PIE System). Given the PIE system $G\{\mathcal{T}, \mathcal{A}, \mathcal{B}, \mathcal{C}, \mathcal{D}\}$ with PI operators $\mathcal{T}, \mathcal{A}, \mathcal{B}, \mathcal{C}$, and $\mathcal{D}$ of compatible dimensions, and signals $w(t) \in \mathbb{R}L_2^{n_w}[a, b]$, we say $\mathbf{x}(t) \in \mathbb{R}L_2[a, b]^{n_x}$ and $z(t) \in \mathbb{R}L_2^{n_z}[a, b]$ satisfy the PIE with initial condition $\mathbf{x}_0$ if

$$\begin{bmatrix} \partial_t(\mathcal{T}\mathbf{x})(t) \\ z(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A} & \mathcal{B} \\ \mathcal{C} & \mathcal{D} \end{bmatrix} \begin{bmatrix} \mathbf{x}(t) \\ w(t) \end{bmatrix}$$

for all $t \geq 0$ and $\mathcal{T}\mathbf{x}(0) = \mathcal{T}\mathbf{x}_0$. Additionally, for signals $\underline{w}(t) \in \mathbb{R}L_2[a, b]^{n_z}$ of compatible dimensions, we say $\underline{\mathbf{x}}(t) \in$

$\mathbb{R}L_2[a, b]^{n_x}$ and $\underline{z}(t) \in L_2[a, b]^{n_w}$ satisfy the dual PIE, $G^\top$, with initial condition $\underline{\mathbf{x}}_0$ if

$$\begin{bmatrix} \mathcal{T}^*\underline{\mathbf{x}}(t) \\ \underline{z}(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}^* & \mathcal{C}^* \\ \mathcal{B}^* & \mathcal{D}^* \end{bmatrix} \begin{bmatrix} \underline{\mathbf{x}}(t) \\ \underline{w}(t) \end{bmatrix},$$

for all $t \geq 0$ and $\mathcal{T}^*\underline{\mathbf{x}}(0) = \mathcal{T}^*\underline{\mathbf{x}}_0$.

Definition 5 (Bounded Internally Stable PIE). Given a PIE system G . We say G is input-output stable if for any $z, w \in \mathbf{L}_{e,[a,b]}$ satisfying $z = Gw$ with $x_0 = 0$, we have

$$\|p_T z\|_{\mathbf{L}_{[a,b]}} \leq \gamma \|p_T w\|_{\mathbf{L}_{[a,b]}}, \quad \forall T \geq 0, \quad \forall w \in \mathbf{L}_{e,[a,b]}^{n_w}.$$

Definition 6 (Redheffer Star Product). For compatible causal systems or operators G and H , write

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = G \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}.$$

The Redheffer star products [19] denote the feedback maps

$$\begin{aligned} G \star H : u_2 &\mapsto y_2, & u_1 &= Hy_1, \\ H \star G : u_1 &\mapsto y_1, & u_2 &= Hy_2, \end{aligned}$$

provided the respective interconnections are well-posed. Here, $P \star \Delta$ closes the uncertainty loop and $K \star P$ closes the controller loop.

C. PIETOOLS

PIETOOLS is a MATLAB toolbox used for declaration, manipulation and solving LPIs [24]. Its operator syntax, PDE-to-PIE conversion, and optimization workflows are documented in the user manual [27]; see also the PIETOOLS homepage.

III. IQC THEORY

This section introduces the IQC objects used to study stability and performance of interconnected systems in a Hilbert space. The construction builds on the IQC framework of Megretski and Rantzer [11] and its extension to hard IQCs for infinite-dimensional channels [29]. We establish the generalized system interconnection, formalize stability and performance, and define the uncertainty and performance IQCs. These definitions provide the foundation for the primal-system analysis in Section IV.

A. Description of uncertain systems in a Hilbert-space

We consider operators that map external inputs to external outputs through a feedback interconnection between a linear (infinite dimensional) operator P and Δ . This interconnection structure is formally defined as follows

Definition 7 (Generalized Plant). Given operators $P_\Delta : \mathbf{L}_{e,[a,b]}^{n_{w\Delta}} \rightarrow \mathbf{L}_{e,[a,b]}^{n_{z\Delta}}$, $P_{\Delta p} : \mathbf{L}_{e,[a,b]}^{n_{wp}} \rightarrow \mathbf{L}_{e,[a,b]}^{n_{z\Delta}}$, $P_{p\Delta} : \mathbf{L}_{e,[a,b]}^{n_{w\Delta}} \rightarrow \mathbf{L}_{e,[a,b]}^{n_{zp}}$, $P_p : \mathbf{L}_{e,[a,b]}^{n_{wp}} \rightarrow \mathbf{L}_{e,[a,b]}^{n_{zp}}$, and $\Delta : \mathbf{L}_{e,[a,b]}^{n_{z\Delta}} \rightarrow \mathbf{L}_{e,[a,b]}^{n_{w\Delta}}$, define

$$\begin{bmatrix} z_\Delta \\ z_p \end{bmatrix} = P \begin{bmatrix} w_\Delta \\ w_p \end{bmatrix} = \begin{bmatrix} P_\Delta & P_{\Delta p} \\ P_{p\Delta} & P_p \end{bmatrix} \begin{bmatrix} w_\Delta \\ w_p \end{bmatrix}, \quad w_\Delta = \Delta z_\Delta. \quad (1)$$

for input $w_p \in \mathbf{L}_{e,[a,b]}^{n_{w_p}}$ and output $z_p \in \mathbf{L}_{e,[a,b]}^{n_{z_p}}$, there exist internal signals $z_\Delta \in \mathbf{L}_{e,[a,b]}^{n_{z_\Delta}}$ and $w_\Delta \in \mathbf{L}_{e,[a,b]}^{n_{w_\Delta}}$ satisfying the interconnected system $z_p = (P \star \Delta)w_p$.

The interconnection structure described above captures the feedback interaction between the nominal system P and the uncertainty Δ , where w_p and z_p represent external inputs and performance outputs, respectively, while w_Δ and z_Δ form the internal feedback loop through the uncertainty. Before analyzing stability and performance, we must first ensure the well-posedness of the interconnection, such that solutions exist, are unique, and depend causally on inputs.

Definition 8. We say that the uncertain interconnection $P \star \Delta$ is well-posed if:

- Existence and Uniqueness, For any $w \in \mathbf{L}_{e,[a,b]}$, there exist unique $z \in \mathbf{L}_{e,[a,b]}$, $z_\Delta \in \mathbf{L}_{e,[a,b]}$ and $w_\Delta \in \mathbf{L}_{e,[a,b]}$ such that, z , w , z_Δ , w_Δ satisfy the interconnection.
- Causality, If z_Δ , w_Δ and $\hat{z}_\Delta$, $\hat{w}_\Delta$ satisfy the interconnection, then $p_T(z_\Delta - \hat{z}_\Delta) = 0$ and $p_T(w_\Delta - \hat{w}_\Delta) = 0$, for all $T \geq 0$.

Having established the class of uncertain systems and their well-posedness, we now develop a framework to guarantee robust stability and performance for the class of systems defined in Definition 7.

B. Robust stability and performance

IQCs describe both uncertainty and performance specifications through quadratic constraints on interconnection signals [34]. For PIE systems, this formulation also supports structured robustness and performance analysis [10]. We now define what it means for the system to remain stable despite uncertainties. Robust stability requires that the system is well-posed for all admissible uncertainties and that the input-output map remains bounded. This involves removal of the performance channels in (1), and introducing auxiliary signals z_0 and w_0 .

Definition 9. The interconnection

$$z_\Delta = P_\Delta w_\Delta + z_0, \quad w_\Delta = \Delta z_\Delta + w_0,$$

with $z_0, w_0 \in \mathbf{L}_{e,[a,b]}$, is said to be robustly stable if:

- 1) The uncertain system is well-posed for all $\Delta \in \Delta$.
- 2) The map from $\begin{bmatrix} z_0 \\ w_0 \end{bmatrix}$ to $\begin{bmatrix} z_\Delta \\ w_\Delta \end{bmatrix}$ is bounded on $\mathbf{L}_{e,[a,b]}$ for all $\Delta \in \Delta$.

Within the IQC framework, the uncertainty operator Δ is described by a quadratic constraint defined by an appropriate multiplier. A stability certificate combines this constraint with a complementary condition on the plant. We use the finite-horizon, or hard, IQC formulation [23], [29], leading to the following definition.

Definition 10. An uncertainty operator $\Delta : \mathbf{L}_{e,[a,b]}^{n_{z_\Delta}} \rightarrow \mathbf{L}_{e,[a,b]}^{n_{w_\Delta}}$ is said to satisfy the hard IQC $(\Psi_\Delta, \mathcal{M}_{\Psi_\Delta})$ defined by the dynamic operator $\Psi_\Delta : \mathbf{L}_{e,[a,b]}^{n_{z_\Delta} + n_{w_\Delta}} \rightarrow \mathbf{L}_{e,[a,b]}^{n_{z_\Delta} + n_{w_\Delta}}$ and $\mathcal{V}_\Delta =$

$\mathcal{V}_\Delta^* \in \mathcal{L}(\mathbb{R}L_2, \mathbb{R}L_2)$ if the inequality holds for all $z_\Delta \in \mathbf{L}_{e,[a,b]}$ and all $T \geq 0$,

$$\left\langle p_T \Psi_\Delta \begin{bmatrix} z_\Delta \\ \Delta z_\Delta \end{bmatrix}, p_T \mathcal{M}_{\Psi_\Delta} \Psi_\Delta \begin{bmatrix} z_\Delta \\ \Delta z_\Delta \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \geq 0.$$

Moreover, let $\mathcal{V}_\Delta \{Q_\Delta, S_\Delta, R_\Delta\}$ be parameterized as,

$$\mathcal{V}_\Delta = \begin{bmatrix} Q_\Delta & S_\Delta \\ S_\Delta^* & R_\Delta \end{bmatrix}$$

Beyond stability, we are interested in ensuring that the system meets certain performance specifications in the presence of uncertainty. Robust performance extends the notion of robust stability by additionally requiring that a performance metric is satisfied.

Definition 11. Robust performance is achieved if the interconnection $P \star \Delta$ is robustly stable and for IQC $(\Psi_p, \mathcal{M}_{\Psi_p})$ defined by $\Psi_p : \mathbf{L}_{e,[a,b]}^{n_{z_p} + n_{w_p}} \rightarrow \mathbf{L}_{e,[a,b]}^{n_{z_p} + n_{w_p}}$ and $\mathcal{V}_p = \mathcal{V}_p^* \in \mathcal{L}(\mathbb{R}L_2, \mathbb{R}L_2)$ we have

$$\left\langle p_T \Psi_p \begin{bmatrix} P \star \Delta \\ I \end{bmatrix} w_p, p_T \mathcal{M}_{\Psi_p} \Psi_p \begin{bmatrix} P \star \Delta \\ I \end{bmatrix} w_p \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon \|p_T w_p\|_{\mathbf{L}_{[a,b]}}^2.$$

for all $w_p \in \mathbf{L}_{e,[a,b]}^{n_{w_p}}$ and $T \geq 0$.

In this framework, the multipliers $\mathcal{M}_{\Psi_\Delta}$ and $\mathcal{M}_{\Psi_p}$ serve distinct but complementary roles. The hard uncertainty IQC $(\Psi_\Delta, \mathcal{M}_{\Psi_\Delta})$ characterizes the behavior of uncertainties, nonlinearities, and other perturbations captured by the operator Δ , ensuring that the IQC is satisfied for the class of admissible disturbances. In contrast, the hard performance IQC $(\Psi_p, \mathcal{M}_{\Psi_p})$ encodes performance requirements for the nominal system, such as L_2 -gain, passivity conditions, or other desired input-output properties. Together, these multipliers enable simultaneous verification of robust stability against perturbations and achievement of specified performance objectives. This section applies the IQC framework to the primal system and gives a certificate for simultaneous robust stability and performance. For brevity, we write the hard IQC as

$$\left\langle p_T \Psi \begin{bmatrix} Pw \\ w \end{bmatrix}, p_T \mathcal{M}_\Psi \Psi \begin{bmatrix} Pw \\ w \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon \|p_T w\|_{\mathbf{L}_{[a,b]}}^2.$$

Let Ψ_Δ and Ψ_p have a 2×2 block partition and define $\Psi_{ij} := \text{diag}((\Psi_\Delta)_{ij}, (\Psi_p)_{ij})$ for $i, j \in \{1, 2\}$. Let $w^\top = [w_\Delta^\top \ w_p^\top]$ and $z^\top = [z_\Delta^\top \ z_p^\top]$. Moreover, define $Q = \text{diag}(Q_\Delta, Q_p)$, $S = \text{diag}(S_\Delta, S_p)$, and $R = \text{diag}(R_\Delta, R_p)$, then $\mathcal{V}\{Q, S, R\}$.

Theorem 1 (Robust Stability and Performance). Let P , Δ be given, and there exists Ψ_Δ , $\mathcal{V}_\Delta$, such that,

- The interconnection $P \star \Delta$ is well-posed for all $\Delta \in \Delta$
- Any uncertainty $\Delta \in \Delta$ satisfy the hard IQC defined by Ψ_Δ and $\mathcal{V}_\Delta$.

Then the interconnection $P \star \Delta$ is robustly stable and robust performance is achieved with respect to some Ψ_p and $\mathcal{V}_p = \{\mathcal{Q}_p, \mathcal{S}_p, \mathcal{R}_p\}$ (Definition 11), with $\mathcal{Q}_p \succeq 0$, if,

$$\left\langle p_T \Psi \begin{bmatrix} Pw \\ w \end{bmatrix}, p_T \mathcal{M}_\mathcal{V} \Psi \begin{bmatrix} Pw \\ w \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon \|p_T w\|_{\mathbf{L}_{[a,b]}}^2 \quad (2)$$

for all $w_\Delta \in \mathbf{L}_{e,[a,b]}^{n_w \Delta}$, $w_p \in \mathbf{L}_{e,[a,b]}^{n_w p}$ and $T \geq 0$.

Proof. See Appendix A.1. ■

Theorem 1 establishes a framework for deriving Linear PI Inequalities (LPIs), which are used in assessing the robust performance of a system.

IV. CONTROLLER ANALYSIS (PRIMAL SYSTEM)

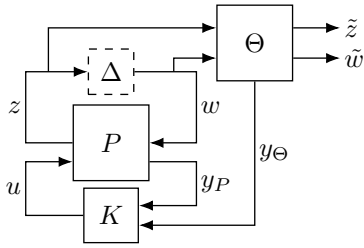


Fig. 2. Primal controller interconnection.

The controller synthesis problem is to find K and an admissible IQC certificate that establish robust stability and performance of the closed-loop interconnection. The primal component block in (3) shows why the primal analysis condition does not directly yield a convex synthesis condition. The controller gains $\mathcal{K}_p$ and $\mathcal{K}_\Theta$ enter both the state dynamics and the filtered outputs. When these gains are decision operators, the storage inequality contains products of the storage operator $\mathcal{P}$ and the controller gains.

For ODE systems, a congruence transformation with $\mathcal{P}^{-1}$, combined with a change of variables for the controller gain multiplied by $\mathcal{P}^{-1}$, removes this bilinearity from the dynamics [41]. For the primal PIE realization, however, the storage operator is preceded by the adjoint of the reconstruction operator. Consequently, the same congruence leaves the factor

$$\mathcal{P}^{-1} \begin{bmatrix} \mathcal{T}_p^* & 0 \\ 0 & \mathcal{T}^* \end{bmatrix} \mathcal{P}.$$

For ODEs the middle factor is the identity, and the storage operators cancel. For PIEs these operators do not generally commute, so this cancellation is unavailable and the transformed inequality remains nonconvex. Thus, although the congruence is well-defined for coercive $\mathcal{P}$, it cannot in general be used to obtain a convex synthesis condition as in the ODE case. Applying an inverse of the reconstruction operator does not generally resolve this issue either, since this inverse need not be bounded on the underlying L_2 space. These structural issues motivate the use of PIE duality for state-feedback synthesis [26].

A second obstacle arises when the controller K and the multiplier weight $\mathcal{V}$ are synthesized simultaneously. In the primal IQC condition, $\mathcal{V}$ weights the quadratic expression in the filtered closed-loop signals. Since these signals depend on K , this expression couples the controller gains and the multiplier weight nonconvexly. The controller-dependent output rows in (3) display this dependence explicitly; using the J -spectral factor Θ fixes the weight to J_{n_z, n_w} , but does not remove the first obstacle. Joint synthesis of K and $\mathcal{V}$ is therefore generally nonconvex in the original primal variables, even for ODE systems, where the reconstruction operators are identities. Obtaining convex synthesis conditions requires additional structure or a suitable reformulation [20], [39]. For general dynamic multipliers, established synthesis methods alternate between controller design and multiplier analysis [35], [37].

These obstacles motivate the dual formulation developed next. By changing the placement of the controller in the component realization, it enables the alternative change of variables used for controller synthesis.

[Placeholder: Specify the class of systems and the assumptions under which the proposed synthesis method applies.]

V. DUALITY

We consider a generalized plant P with a PIE state-space realization. The nominal PIE duality results of Shivakumar et al. [26] provide the starting point for convex state-feedback synthesis. For dynamically filtered IQCs, two further ingredients are relevant: hard factorizations connecting frequency-domain multipliers to finite-horizon inequalities [23], and the dual-filter and storage transformations developed for LPV estimation and feedforward synthesis [36]. This section develops the corresponding dual-filter construction for PIE systems using the Potapov–Ginzburg transform and PIE gain duality.

A. Duality under Dynamical Filtering

We begin by restating the main theorem from [26].

Theorem 2 (Dual L_2 -gain [26, Thm. 9]). *For a PIE system with corresponding PI operators $P\{\mathcal{T}, \mathcal{A}, \mathcal{B}, \mathcal{C}, \mathcal{D}\}$, the following statements are equivalent.*

- 1) *For any $w \in \mathbf{L}_{e,[a,b]}^{n_w}$, and $\mathbf{x}(0) = 0$, P satisfies $\|p_T z\| \leq \|p_T w\|$.*
- 2) *For any $\underline{w} \in \mathbf{L}_{e,[a,b]}^{n_z}$, and $\underline{\mathbf{x}}(0) = 0$, $P^\top$ satisfies $\|p_T \underline{z}\| \leq \|p_T \underline{w}\|$.*

For a bounded, time-invariant, memoryless in time linear uncertainty Δ , suppose the feedback interconnection is well-posed and its closed-loop operators satisfy the hypotheses of [26, Thm. 6]. Closing the loop $w = \Delta z$ gives

$$\mathcal{T}\dot{\mathbf{x}} = (\mathcal{A} + \mathcal{B}\Delta(I - \mathcal{D}\Delta)^{-1}\mathcal{C})\mathbf{x}.$$

Taking adjoints and using the push-through identity yields

$$(\mathcal{T})^* = \mathcal{T}^*, \quad (\mathcal{A} + \mathcal{B}\Delta(I - \mathcal{D}\Delta)^{-1}\mathcal{C})^* = \mathcal{A}^* + \mathcal{C}^*\Delta^*(I - \mathcal{D}^*\Delta^*)^{-1}\mathcal{B}^*.$$

This is exactly the state operator of $P^\top \star \Delta^*$. Thus, [26, Thm. 6] establishes equivalence of weak convergence for

$$\begin{aligned}
P &:= \begin{cases} \begin{bmatrix} \mathcal{T}_P \dot{\mathbf{x}}_P(t) \\ z(t) \\ y_P(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}_P & \mathcal{B}_P & \mathcal{B}_u \\ \mathcal{C}_P & \mathcal{D}_P & \mathcal{D}_{zu} \\ I & 0 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x}_P(t) \\ w(t) \\ u(t) \end{bmatrix} \\ K &:= \begin{cases} u(t) = [\mathcal{K}_P \quad \mathcal{K}_\Theta] \begin{bmatrix} \mathbf{x}_P(t) \\ \mathbf{x}_\Theta(t) \end{bmatrix} \end{cases} \\ \Theta &:= \begin{cases} \begin{bmatrix} \mathcal{T}_\Theta \dot{\mathbf{x}}_\Theta(t) \\ \tilde{v}(t) \\ y_\Theta(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A} & \mathcal{B} \\ \mathcal{C} & \mathcal{D} \\ I & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x}_\Theta(t) \\ v(t) \end{bmatrix} \end{cases} \\ \Theta \begin{bmatrix} K \star P \\ I \end{bmatrix} &:= \begin{cases} \begin{bmatrix} \mathcal{T}_P \dot{\mathbf{x}}_P(t) \\ \mathcal{T}_\Theta \dot{\mathbf{x}}_\Theta(t) \\ \tilde{z}(t) \\ \tilde{w}(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}_P + \mathcal{B}_u \mathcal{K}_P & \mathcal{B}_u \mathcal{K}_\Theta & \mathcal{B}_P \\ \mathcal{B}_1(\mathcal{C}_P + \mathcal{D}_{zu} \mathcal{K}_P) & \mathcal{A} + \mathcal{B}_1 \mathcal{D}_{zu} \mathcal{K}_\Theta & \mathcal{B}_1 \mathcal{D}_P + \mathcal{B}_2 \\ \mathcal{D}_{11}(\mathcal{C}_P + \mathcal{D}_{zu} \mathcal{K}_P) & \mathcal{C}_1 + \mathcal{D}_{11} \mathcal{D}_{zu} \mathcal{K}_\Theta & \mathcal{D}_{11} \mathcal{D}_P + \mathcal{D}_{12} \\ \mathcal{D}_{21}(\mathcal{C}_P + \mathcal{D}_{zu} \mathcal{K}_P) & \mathcal{C}_2 + \mathcal{D}_{21} \mathcal{D}_{zu} \mathcal{K}_\Theta & \mathcal{D}_{21} \mathcal{D}_P + \mathcal{D}_{22} \end{bmatrix} \begin{bmatrix} \mathbf{x}_P(t) \\ \mathbf{x}_\Theta(t) \\ w(t) \end{bmatrix} \end{cases} \end{cases} \quad (3)
\end{aligned}$$

Fig. 1. Primal component realizations and full augmented interconnection for state feedback with dynamic filtering.

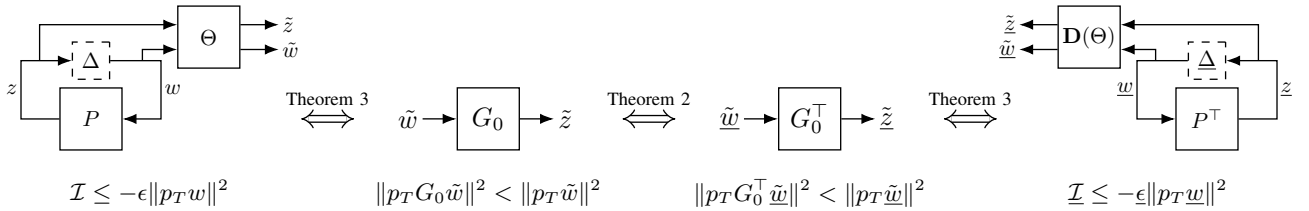


Fig. 3. Equivalence chain for primal and dual dynamic filtering.

the primal and dual states. Exponential stability is likewise equivalent by [26, Cor. 8]. For the constant repeated real uncertainty $\Delta = \delta I$ in Example 1, $\Delta^* = \Delta$, so the same uncertainty appears in both interconnections. For a general uncertainty Δ , this construction is less trivial. It is therefore more convenient to develop duality through the IQC description, whose filters and multiplier weights are linear operators, rather than through Δ itself. The explicit uncertainty-adjoint relation of [8] assumes linear uncertainty. Venkataraman and Seiler [36] accommodate nonlinear uncertainties through dual IQCs and J -spectral factorizations of strict positive-negative multipliers. We adopt the same strategy: first reformulate the IQC condition as a contraction requirement on an associated system G_0 with respect to the truncated L_H norm, then apply the previously stated theorem, and finally recover the original formulation using the dual filter. In their framework, this relationship is established through congruence transformations of the analysis LMIs [36, Sec. 3]. Our approach relies on PIE gain duality together with truncated signal inequalities imposed over arbitrary finite horizons. This provides a natural extension of the methodology to robust feedback synthesis for PDE systems.

Relating an IQC to this gain duality requires two transformations. First, write $\begin{bmatrix} \tilde{z} \\ \tilde{w} \end{bmatrix} := \Theta \begin{bmatrix} z \\ w \end{bmatrix}$. A J -spectral factorization gives, for every $T \geq 0$,

$$\begin{aligned}
&\left\langle p_T \Psi \begin{bmatrix} z \\ w \end{bmatrix}, p_T \mathcal{M}_V \Psi \begin{bmatrix} z \\ w \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \\
&= \|p_T \tilde{z}\|_{\mathbf{L}_{[a,b]}}^2 - \|p_T \tilde{w}\|_{\mathbf{L}_{[a,b]}}^2.
\end{aligned}$$

Thus the IQC quadratic form becomes the difference between

the filtered output and input signals.

Definition 12 (J -Spectral Factorization and Dual Filter). *For a given IQC defined by $(\Psi, \mathcal{V})$, we say that a causal linear bounded operator Θ and $J_{\mathbf{n}_z, \mathbf{n}_w} := \begin{bmatrix} I_{\mathbf{n}_z} & 0 \\ 0 & -I_{\mathbf{n}_w} \end{bmatrix}$ (the pair $(\Theta, J_{\mathbf{n}_z, \mathbf{n}_w})$) is a J -spectral factorization of $(\Psi, \mathcal{V})$ if,*

$$\begin{aligned}
&\left\langle p_T \Psi \begin{bmatrix} z \\ w \end{bmatrix}, p_T \mathcal{M}_V \Psi \begin{bmatrix} z \\ w \end{bmatrix} \right\rangle \\
&= \left\langle p_T \Theta \begin{bmatrix} z \\ w \end{bmatrix}, p_T J_{\mathbf{n}_z, \mathbf{n}_w} \Theta \begin{bmatrix} z \\ w \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}},
\end{aligned}$$

Following the dual-filter construction in [36], if Θ has a causal bounded inverse, we use $\mathbf{D}(\Theta)$ to denote the dual filter associated with Θ of the form

$$\mathbf{D}(\Theta) := (\hat{J}^* \Theta^\top \hat{J})^{-1},$$

$$\text{where } \hat{J} := \begin{bmatrix} 0_{\mathbf{n}_z \times \mathbf{n}_w} & I_{\mathbf{n}_z} \\ -I_{\mathbf{n}_w} & 0_{\mathbf{n}_w \times \mathbf{n}_z} \end{bmatrix}.$$

Give the primal inner product,

$$\mathcal{I} := \left\langle p_T \Theta \begin{bmatrix} Pw \\ w \end{bmatrix}, p_T J_{\mathbf{n}_z, \mathbf{n}_w} \Theta \begin{bmatrix} Pw \\ w \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}}$$

and corresponding dual iq

$$\mathcal{I} := \left\langle p_T \mathbf{D}(\Theta) \begin{bmatrix} P^\top \underline{w} \\ \underline{w} \end{bmatrix}, p_T J_{\mathbf{n}_w, \mathbf{n}_z} \mathbf{D}(\Theta) \begin{bmatrix} P^\top \underline{w} \\ \underline{w} \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}}$$

and $\Delta : \mathbf{L}_{e,[a,b]}^{\mathbf{n}_z \Delta} \rightarrow \mathbf{L}_{e,[a,b]}^{\mathbf{n}_w \Delta}$,

Theorem 3. Let P be as in Definition 7 and let $(\begin{bmatrix} \Theta_{11} & \Theta_{12} \\ \Theta_{21} & \Theta_{22} \end{bmatrix}, J_{\mathbf{n}_z, \mathbf{n}_w})$ be a hard J -spectral factorization with. Finally, assume $(\Theta_{21}P + \Theta_{22})$ has a bounded causal inverse. Define

$$G_0 = (\Theta_{11}P + \Theta_{12})(\Theta_{21}P + \Theta_{22})^{-1}$$

then the following statements are equivalent,

- 1) $\|p_T G_0 \tilde{w}\|_{\mathbf{L}_{[a,b]}}^2 < \|p_T \tilde{w}\|_{\mathbf{L}_{[a,b]}}^2$ for every $\tilde{w}$ and $T \geq 0$.
- 2) There exists some $\epsilon > 0$ such that $\mathcal{I} \leq -\epsilon \|p_T w\|_{\mathbf{L}_{[a,b]}}^2$ for every w and $T \geq 0$.
- 3) There exists some $\underline{\epsilon} > 0$ such that $\underline{\mathcal{I}} \leq -\underline{\epsilon} \|p_T \underline{w}\|_{\mathbf{L}_{[a,b]}}^2$ for every $\underline{w}$ and $T \geq 0$.

Proof. See Appendix A.2. $\blacksquare$

Theorem 4. Let the realization of P satisfy Definition 7. Let $\Delta \in \Delta$ and there exist *either*, a primal hard IQC $(\Psi, \mathcal{M}_\Psi)$, with $\Psi \in \Pi_4$, or a dual hard IQC $(\underline{\Psi}, \mathcal{M}_{\underline{\Psi}})$, with $\underline{\Psi} \in \Pi_4$, such that,

- The interconnection $P \star \Delta$ is well-posed for all $\Delta \in \Delta$.
- The IQC in Definition 10 is satisfied for all $\Delta \in \Delta$.
- The primal hard IQC $(\Psi, \mathcal{M}_\Psi)$ has a hard $(\Theta, J_{\mathbf{n}_z, \mathbf{n}_w})$ factorization.

Suppose that *either* of the two statements hold for some $\mathcal{P} = \mathcal{P}^* \succeq 0 \in \Pi_4$.

- $\begin{bmatrix} \mathcal{T}^* \mathcal{P} \mathcal{A} + (*) & \mathcal{T}^* \mathcal{P} \mathcal{B} \\ \mathcal{B}^* \mathcal{P} \mathcal{T} & \epsilon I \end{bmatrix} + \begin{bmatrix} \mathcal{C}^* \\ \mathcal{D}^* \end{bmatrix} \mathcal{V} \begin{bmatrix} \mathcal{C} & \mathcal{D} \end{bmatrix} \preceq 0$
- $\begin{bmatrix} \mathcal{T}^* \mathcal{P} \mathcal{A} + (*) & \mathcal{T}^* \mathcal{P} \mathcal{B} \\ \underline{\mathcal{B}}^* \mathcal{P} \underline{\mathcal{T}} & \underline{\epsilon} I \end{bmatrix} + \begin{bmatrix} \underline{\mathcal{C}}^* \\ \underline{\mathcal{D}}^* \end{bmatrix} \underline{\mathcal{V}} \begin{bmatrix} \underline{\mathcal{C}} & \underline{\mathcal{D}} \end{bmatrix} \preceq 0$ (4)

Then for any $w, \underline{w} \in \mathbf{L}_{e,[a,b]}$, satisfies either

$$\Psi \begin{bmatrix} P \\ I \end{bmatrix} w := \left\{ \begin{bmatrix} \mathcal{T} \dot{\mathbf{x}}(t) \\ v(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A} & \mathcal{B} \\ \mathcal{C} & \mathcal{D} \end{bmatrix} \begin{bmatrix} \mathbf{x}(t) \\ w(t) \end{bmatrix} \right\}$$

or

$$\underline{\Psi} \begin{bmatrix} P^\top \\ I \end{bmatrix} \underline{w} := \left\{ \begin{bmatrix} \mathcal{T} \dot{\underline{\mathbf{x}}}(t) \\ \underline{v}(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A} & \mathcal{B} \\ \underline{\mathcal{C}} & \underline{\mathcal{D}} \end{bmatrix} \begin{bmatrix} \underline{\mathbf{x}}(t) \\ \underline{w}(t) \end{bmatrix} \right\} \quad (5)$$

Then the interconnection $P \star \Delta$ with $\mathbf{x}(0) = 0$ is robustly performing.

Proof. See Appendix A.3. $\blacksquare$

Remark 1. Note that we do not require information about the dual uncertainty $\underline{\Delta}$, only the primal uncertainty Δ .

VI. CONVEX CONTROLLER SYNTHESIS (DUAL SYSTEM)

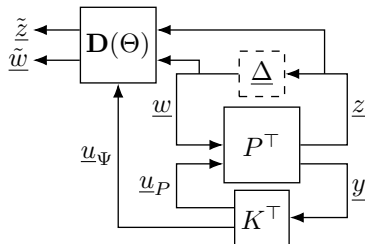


Fig. 5. Dual controller interconnection.

Theorem 4 provides a robust-performance certificate for PIE systems in terms of the dual IQC. For analysis, this merely reformulates the primal certificate, since primal–dual equivalence preserves feasibility. The benefit of the dual formulation therefore appears in controller synthesis. Both controller constructions below use the same synthesis LPI, obtained by applying (4) to the corresponding closed-loop PIE. The choice of dual uncertainty filter changes only the realization of the controller recovered from this LPI. Related applications of IQC duality include robust estimation [21] and disturbance-feedforward controller synthesis [8], [36].

A. Primal and Dual Systems

Let P , with zero initial condition $\mathbf{x}(0) = 0$, have input w , control input u , and output z . Its primal and dual PIE state-space realizations are summarized in the corresponding system-component boxes. If $\Theta \neq I$, assume that z, w are measurable.

B. State Feedback LPI

Corollary 1. Let the realization of P satisfy Definition 7. Let $\Delta \in \Delta$ and suppose a primal hard IQC has the hard $(\Theta, J_{\mathbf{n}_z, \mathbf{n}_w})$ factorization. Assume that

- The interconnection $P \star \Delta$ is well-posed for all $\Delta \in \Delta$.
- The IQC in Definition 10 is satisfied for all $\Delta \in \Delta$.

Moreover there exist, $\mathcal{Z}, \mathcal{P} \in \Pi_4$ with $\mathcal{P} = \mathcal{P}^* \succ 0$, such that the following statement holds

$$\begin{bmatrix} \mathcal{T}^* \mathcal{P} \mathcal{A}_0 + \mathcal{T}^* \mathcal{Z} \begin{bmatrix} \mathcal{B}_u^* & 0 \end{bmatrix} + (*) & \mathcal{T}^* \mathcal{P} \mathcal{B}_0 + \mathcal{T}^* \mathcal{Z} \mathcal{D}_{zu}^* \\ (*) & \underline{\epsilon} I \end{bmatrix} + \begin{bmatrix} \mathcal{C}^* \\ \underline{\mathcal{D}}^* \end{bmatrix} J_{\mathbf{n}_w, \mathbf{n}_z} \begin{bmatrix} \underline{\mathcal{C}} & \underline{\mathcal{D}} \end{bmatrix} \preceq 0. \quad (7)$$

where

$$\mathbf{D}(\Theta) \begin{bmatrix} P^\top \\ I \end{bmatrix} \underline{w} := \left\{ \begin{bmatrix} \mathcal{T} \dot{\underline{\mathbf{x}}}(t) \\ \underline{v}(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}_0 & \mathcal{B}_0 \\ \underline{\mathcal{C}} & \underline{\mathcal{D}} \end{bmatrix} \begin{bmatrix} \underline{\mathbf{x}}(t) \\ \underline{w}(t) \end{bmatrix} \right\} \quad (8)$$

and recover the controller gains from

$$\begin{bmatrix} \mathcal{K}_P & \mathcal{K}_\Theta \end{bmatrix} = \mathcal{Z}^* \mathcal{P}^{-1}.$$

Assume that $\mathcal{D}_{21} \mathcal{D}_P + \mathcal{D}_{22}$ is invertible, with

$$\Lambda := (\mathcal{D}_{21} \mathcal{D}_P + \mathcal{D}_{22})^{-1} \in \Pi_4$$

Then the interconnection $K \star P \star \Delta$ with $\mathbf{x}(0) = 0$ is robustly stable and robust performance is achieved.

C. $\underline{\Psi}_\Delta = I$ controller realization

When $\Psi_\Delta = I$ and $\underline{\Psi}_\Delta = I$, no multiplier state is required in the controller. For a state-feedback operator $\mathcal{K}_P$, set $u(t) = \mathcal{K}_P \mathbf{x}_P(t)$. The resulting closed-loop PIE $\mathcal{K} \star \mathcal{P}$ is

$$\begin{bmatrix} \partial_t (\mathcal{T}_P \mathbf{x}_P)(t) \\ z(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}_P + \mathcal{B}_u \mathcal{K}_P & \mathcal{B}_P \\ \mathcal{C}_P + \mathcal{D}_{zu} \mathcal{K}_P & \mathcal{D}_P \end{bmatrix} \begin{bmatrix} \mathbf{x}_P(t) \\ w(t) \end{bmatrix}.$$

$$\begin{aligned}
P^\top &:= \left\{ \begin{bmatrix} \mathcal{T}_P^* \dot{\mathbf{x}}_P(t) \\ \underline{z}(t) \\ \underline{y}(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}_P^* & \mathcal{C}_P^* & I \\ \mathcal{B}_P^* & \mathcal{D}_P^* & 0 \\ \mathcal{B}_u^* & \mathcal{D}_{zu}^* & 0 \end{bmatrix} \begin{bmatrix} \underline{\mathbf{x}}_P(t) \\ \underline{w}(t) \\ \underline{u}_P(t) \end{bmatrix} \right. \\
K^\top &:= \left\{ \begin{bmatrix} \underline{u}_P(t) \\ \underline{u}_\Theta(t) \end{bmatrix} = \begin{bmatrix} \mathcal{K}_P^* \\ \mathcal{K}_\Theta^* \end{bmatrix} y(t) \right. \\
\mathbf{D}(\Theta) &:= \left\{ \begin{bmatrix} \mathcal{T}^* \dot{\mathbf{x}}_\Theta(t) \\ \underline{v}(t) \\ \underline{u}_\Theta(t) \end{bmatrix} = \begin{bmatrix} \underline{\mathcal{A}} & \underline{\mathcal{B}} & I \\ \underline{\mathcal{C}} & \underline{\mathcal{D}} & 0 \end{bmatrix} \begin{bmatrix} \underline{\mathbf{x}}_\Theta(t) \\ \underline{v}(t) \\ \underline{u}_\Theta(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}^* - \mathcal{C}^* \hat{\mathbf{J}} \mathbf{D}(\mathcal{D}) \hat{\mathbf{J}}^* \mathcal{B}^*, & \mathcal{C}^* \hat{\mathbf{J}} \mathbf{D}(\mathcal{D}) & I \\ -\mathbf{D}(\mathcal{D}) \hat{\mathbf{J}}^* \mathcal{B}^*, & \mathbf{D}(\mathcal{D}) & 0 \end{bmatrix} \begin{bmatrix} \underline{\mathbf{x}}_\Theta(t) \\ \underline{v}(t) \\ \underline{u}_\Theta(t) \end{bmatrix} \right. \\
\mathbf{D}(\Theta) \begin{bmatrix} (K \star P)^\top \\ I \end{bmatrix} &:= \left\{ \begin{bmatrix} \mathcal{T}_P^* \dot{\mathbf{x}}_P(t) \\ \mathcal{T}^* \dot{\mathbf{x}}_\Theta(t) \\ \underline{\tilde{z}}(t) \\ \underline{\tilde{w}}(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}_P^* + \mathcal{K}_P^* \mathcal{B}_u^* & 0 & \mathcal{C}_P^* + \mathcal{K}_P^* \mathcal{D}_{zu}^* \\ \underline{\mathcal{B}}_1 \mathcal{B}_P^* + \mathcal{K}_\Theta^* \mathcal{B}_u^* & \underline{\mathcal{A}} & \underline{\mathcal{B}}_1 \mathcal{D}_P^* + \underline{\mathcal{B}}_2 + \mathcal{K}_\Theta^* \mathcal{D}_{zu}^* \\ \underline{\mathcal{D}}_{11} \mathcal{B}_P^* & \underline{\mathcal{C}}_1 & \underline{\mathcal{D}}_{11} \mathcal{D}_P^* + \underline{\mathcal{D}}_{12} \\ \underline{\mathcal{D}}_{21} \mathcal{B}_P^* & \underline{\mathcal{C}}_2 & \underline{\mathcal{D}}_{21} \mathcal{D}_P^* + \underline{\mathcal{D}}_{22} \end{bmatrix} \begin{bmatrix} \underline{\mathbf{x}}_P(t) \\ \underline{\mathbf{x}}_\Theta(t) \\ \underline{w}(t) \end{bmatrix} \right. \quad (6)
\end{aligned}$$

Fig. 4. Dual component realizations and full augmented interconnection for state feedback with dynamic filtering, expressed in terms of the primal PI operators.

D. Dynamic-filter controller realization

Dynamic-IQC feedback constructions with additional controller memory have also been studied for uncertain systems with state delay [39]. Here, assume that the uncertainty Δ is known and that z_Δ , $w_\Delta = \Delta z_\Delta$, and w_p are available to the controller. For $\Psi_\Delta \neq I$, find the realization of $\Psi_\Delta = \mathbf{D}(\Psi_\Delta)$,

$$\begin{bmatrix} \mathcal{T}_{\Psi_\Delta} \dot{\mathbf{x}}_{\Psi_\Delta}(t) \\ y(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}_{\Psi_\Delta} & \mathcal{B}_{\Psi_\Delta}^z & \mathcal{B}_{\Psi_\Delta}^w \\ I & 0 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x}_{\Psi_\Delta}(t) \\ z_\Delta(t) \\ w_\Delta(t) \end{bmatrix}.$$

We impose the zero initial condition $\mathbf{x}_{\Psi_\Delta}(0) = 0$. The second block row gives $y(t) = \mathbf{x}_{\Psi_\Delta}(t)$. The complete controller has the PIE realization

$$\begin{bmatrix} \mathcal{T}_{\Psi_\Delta} \dot{\mathbf{x}}_{\Psi_\Delta}(t) \\ u(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}_{\Psi_\Delta} & 0 & \mathcal{B}_{\Psi_\Delta}^z & \mathcal{B}_{\Psi_\Delta}^w \\ \mathcal{K}_\Psi & \mathcal{K}_P & 0 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x}_{\Psi_\Delta}(t) \\ \mathbf{x}(t) \\ z_\Delta(t) \\ w_\Delta(t) \end{bmatrix}.$$

Hence $\mathbf{x}_{\Psi_\Delta}$ is the controller state, $\mathbf{x}$, and relevant, z_Δ , and w_Δ are its external inputs, and u is its output.

VII. NUMERICAL EXAMPLES

We will consider examples from [26] but add uncertainty.

Example 1 (Reaction-Diffusion Equation). Consider boundary control of an unstable uncertain reaction-diffusion equation:

$$\begin{aligned}
\dot{\mathbf{v}}(t, s) &= \mathbf{v}_{ss}(t, s) + 5\mathbf{v}(t, s) + 10w_{\Delta_1}(t, s) + sw_p(t), \\
\mathbf{v}(t, 0) &= 0, \quad \mathbf{v}_s(t, 1) = x(t), \quad \dot{x}(t) = u(t), \\
z_{\Delta_1}(t, s) &= \mathbf{v}(t, s), \quad (\Delta_1 z_{\Delta_1})(t, s) = \delta z_{\Delta_1}(t, s), \\
w_{\Delta_1} &= \Delta_1 z_{\Delta_1}, \quad |\delta| < 1, \quad z_p(t) = \left[\int_0^1 \mathbf{v}(t, s) ds \right].
\end{aligned}$$

The PIE realization of this primal representation has state $\mathbf{x}(t) = [x(t) \quad \mathbf{v}_{ss}(t)]^\top$. Define the spatial integration opera-

tor $\mathcal{J} \in \Pi_3$ by $(\mathcal{J}f)(t, s) := \int_0^s f(t, \theta) d\theta$, and introduce the filtered uncertainty channel

$$\begin{aligned}
\dot{\mathbf{v}}(t, s) &= \mathbf{v}_{ss}(t, s) + 5\mathbf{v}(t, s) + 10(\mathcal{J}w_{\Delta_2})(t, s) + sw_p(t), \\
\mathbf{v}(t, 0) &= 0, \quad \mathbf{v}_s(t, 1) = x(t), \quad \dot{x}(t) = u(t), \\
z_{\Delta_2}(t, s) &= \mathbf{v}(t, s), \quad (\Delta_2 z_{\Delta_2})(t, s) = \delta z_{\Delta_2}(t, s), \\
w_{\Delta_2} &= \Delta_2 z_{\Delta_2}, \quad |\delta| < 1, \quad z_p(t) = \left[\int_0^1 \mathbf{v}(t, s) ds \right].
\end{aligned}$$

Note that $sw_p(t) = (\mathcal{J}w_p)(t, s)$. This filtering produces an equivalent plant because the same scalar δ acts at every spatial location and $\mathbf{v}(t, 0) = 0$:

$$(\mathcal{J}w_{\Delta_2})(t, s) = \int_0^s \delta \mathbf{v}_\theta(t, \theta) d\theta = \delta \mathbf{v}(t, s) = w_{\Delta_1}(t, s).$$

The lifting does not change the uncertainty class, $\Delta_2 = \delta I$ is still a repeated real parametric uncertainty, with the same scalar δ . In particular, $\underline{\Delta}_2 = \Delta_2^* = \Delta_2$, so the same multiplier structure can be used on the primal and dual sides. Following [29], we parameterize the primal multiplier as

$$\mathcal{V}_\Delta \{ \mathcal{Q}_\Delta, \mathcal{S}_\Delta, -\mathcal{Q}_\Delta \},$$

where $\mathcal{Q}_\Delta^* = \mathcal{Q}_\Delta \succeq 0$ and $\mathcal{S}_\Delta^* = -\mathcal{S}_\Delta$, with $\mathcal{Q}_\Delta, \mathcal{S}_\Delta \in \Pi_4$ and $\Psi_\Delta = I$. The corresponding dual multiplier is parameterized analogously.

For the induced L_2 -gain from w_p to z_p , define the primal performance multiplier $\mathcal{V}_p \{ I_2, 0, -\gamma^2 \}$ and the dual performance multiplier $\underline{\mathcal{V}}_p \{ 1, 0, -\gamma^2 I_2 \}$. Moreover, the control gain is also bounded $\|\mathcal{K}\|_{\mathcal{L}(\mathbf{L})} \leq 10^4$ as an LPI constraint. Synthesis using this repeated-parametric IQC and Theorem 1 yields

$$\begin{aligned}
u(t) &= -247.26x(t) + \int_0^1 K(s) \mathbf{v}_{ss}(t, s) ds, \\
K(s) &= 1.93s^{10} - 20.76s^9 + 81.61s^8 - 165.38s^7 \\
&\quad + 191.14s^6 - 94.18s^5 + 0.20s^4 - 102.86s^3 \\
&\quad - 14.63s^2 + 335.30s - 0.004
\end{aligned}$$

Synthesis yielded a gain of 0.798, while primal and dual analysis yielded $\gamma = 0.743$ and $\underline{\gamma} = 0.748$, respectively, for the closed-loop interconnection $\bar{K} \star P \star \Delta$.

A. Sector and Slope restricted uncertainty

Lemma 1 (Dual Sector Bound IQC). *Suppose that for $z_\Delta \in \mathbf{L}_e, [a, b]$, $(\Delta z_\Delta)(s, t) = \phi(z_\Delta(s, t))$ for all $s \in [a, b]$ and $t > 0$, where ϕ satisfies*

$$\alpha z^2 \leq z\phi(z) < \beta z^2 \quad \text{for all } z \in \mathbb{R}$$

with $\alpha, \beta \in \mathbb{R}$. Then Δ satisfies the dual Hard IQC defined by $\underline{\Psi}_\Delta = I$ and

$$\underline{\mathcal{V}}_\Delta = \begin{bmatrix} \beta & -1 \\ -\alpha & 1 \end{bmatrix}^* \begin{bmatrix} 0 & \underline{\mathcal{S}}_\Delta \\ \underline{\mathcal{S}}_\Delta & 0 \end{bmatrix} \begin{bmatrix} \beta & -1 \\ -\alpha & 1 \end{bmatrix} \quad (9)$$

for and $\underline{\mathcal{S}}_\Delta \in \Pi_3$ with

$$(\underline{\mathcal{S}}_\Delta \mathbf{x})(s, t) := \underline{\mathcal{S}}_0(s) \mathbf{x}(s, t), \quad \underline{\mathcal{S}}_0(s) > 0. \quad (10)$$

Proof. See Appendix A.5. $\blacksquare$

Optimal and robust boundary control of the sine–Gordon equation has also been studied on finite horizons [16].

Example 2 (Sine–Gordon Wave Equation). *Consider boundary control of the sine–Gordon equation shifted about the unstable equilibrium. Using the spatial integration operator $\mathcal{J}$ introduced in Example 1, the system supplied to the dual graph construction and the controller-synthesis LPI is*

$$\begin{aligned} \ddot{v}(t, s) &= v_{ss}(t, s) - 2\dot{v}(t, s) \\ &\quad + 3(\mathcal{J}w_{\Delta_2})(t, s) + sw_p(t), \\ v(t, 0) &= 0, \quad v_s(t, 1) = x(t), \quad \dot{x}(t) = u(t), \\ z_{\Delta_2}(t, s) &= v_s(t, s), \\ (\Delta_2 z_{\Delta_2})(t, s) &= \cos\left(\int_0^s z_{\Delta_2}(t, \theta) d\theta\right) z_{\Delta_2}(t, s), \\ w_{\Delta_2} &= \Delta_2 z_{\Delta_2}, \quad z_p(t) = \int_0^1 v(t, s) ds. \end{aligned}$$

Note that $sw_p(t) = (\mathcal{J}w_p)(t, s)$. The filtered uncertainty output exactly recovers the sine nonlinearity because

$$\mathcal{J}w_{\Delta_2} = \mathcal{J}(\cos(v)v_s) = \sin(v) - \sin(v(t, 0)) = \sin(v).$$

Furthermore,

$$-z_{\Delta_2}^2 \leq z_{\Delta_2} w_{\Delta_2} \leq z_{\Delta_2}^2,$$

so Δ_2 lies in the sector $[-1, 1]$. Lemma 1 therefore applies with $\alpha = -1$ and $\beta = 1$. We use the corresponding dual sector multiplier $\underline{\mathcal{V}}_\Delta \{0, \underline{\mathcal{S}}_\Delta, 0\}$ from (9), with $\underline{\mathcal{S}}_\Delta$ parameterized as in (10). To bound the induced L_2 -gain from w_p to z_p , we choose the dual performance multiplier $\underline{\mathcal{V}}_p = \{1, 0, -\underline{\gamma}^2\}$ and identify the primal and dual gain bounds through $\gamma = \underline{\gamma}$. We also impose $\|\mathcal{K}\|_{\mathcal{L}(\mathbf{L})} \leq 10^4$ in the LPI formulation to limit the gain of the controller. Solving (7) yields the certified upper bound $\gamma = 0.45$ on the induced L_2 -gain from w_p to z_p of the closed-loop interconnection $K \star P \star \Delta$. The optimized state-feedback controller is

$$u(t) = -15.46x(t) + \int_0^1 Q_1(s) 2v_{ss}(t, s) + Q_2(s) \dot{v}(t, s) ds$$

where

$$\begin{aligned} Q_1(s) &= 200.61s^{12} - 1537.17s^{11} + 5033.27s^{10} - 9096.06s^9 \\ &\quad + 9933.67s^8 - 6788.31s^7 + 2916.66s^6 - 700.58s^5 \\ &\quad - 82.97s^4 + 161.05s^3 - 40.98s^2 + 3.72s - 7.6113 \\ Q_2(s) &= 203.29s^{12} - 1309.04s^{11} + 3721.08s^{10} - 5890.06s^9 \\ &\quad + 5519.18s^8 - 3072.06s^7 + 942.76s^6 + 1.51s^5 \\ &\quad - 274.86s^4 + 166.68s^3 - 21.49s^2 + 11.76s - 0.07 \end{aligned}$$

VIII. CONNECTION TO FINITE-DIMENSIONAL DYNAMIC MULTIPLIERS

This section connects the duality framework developed above with the established theory of finite-dimensional dynamic IQC multipliers. For rational multipliers satisfying the strict positive–negative (PN) conditions, Seiler [23] showed that a J -spectral factorization provides a stable and stably invertible hard factorization, thereby converting the associated soft, infinite-horizon IQC into a time-domain constraint that holds on every finite horizon. As discussed in the linear parameter-varying (LPV) setting by Venkataraman and Seiler [36], this J -spectral factorization satisfies the filter assumptions needed to apply the duality theory. The first part of this section reviews this connection between strict PN multipliers, hard factorizations, and duality. The second part illustrates the construction using a delayed firing-rate model.

A. Positive–Negative Multipliers

Using the terminology of Definition 12, we denote the J -spectral factor by Θ . The following result restates [23, Thm. 4] in this notation. Moreover, for finite dimensional multipliers we can define $\Pi = \Psi \sim V \Psi \in \mathbb{RL}_\infty$, where $\Psi(j\omega)^\sim = \Psi(-j\omega)^\top$.

Theorem 5. *Let $\Pi = \Pi^\sim \in \mathbb{RL}_\infty^{(n_z+n_w) \times (n_z+n_w)}$ where $\Pi_{11} \in \mathbb{RL}_\infty^{n_z \times n_z}$ and $\Pi_{22} \in \mathbb{RL}_\infty^{n_w \times n_w}$. If $\Pi_{11}(j\omega) \succ 0, \Pi_{22}(j\omega) \prec 0$, for all $\omega \in \mathbb{R} \cup \{\infty\}$, then*

- 1) Π admits a J_{n_z, n_w} -spectral factorization $\Pi = \Theta^\sim J_{n_z, n_w} \Theta$, where $\Theta, \Theta^{-1} \in \mathbb{RH}_\infty^{(n_z+n_w) \times (n_z+n_w)}$;
- 2) (Θ, J_{n_z, n_w}) is a hard factorization of Π .

A Riccati-based construction of J -spectral factors for strict PN multipliers is given in [17, App. B], using rational J -spectral factorization results [12].

The distinction between a frequency-domain factorization identity and the finite-horizon conditions used in Definition 12 is relevant here. Preservation of both graph and inverse-graph inequalities is studied through doubly-hard factorizations [1]. Finite-horizon IQCs with terminal costs provide another connection to dissipativity [22], while multiplier storage can reduce conservatism [7], [18]. These results provide context for the finite-horizon assumptions imposed in Definition 12; they do not by themselves establish the truncated quadratic-form identity in that definition.

B. Example: Generation of β -Oscillations in the Subthalamic Nucleus-Globus Pallidus Network

Consider the delayed firing-rate model in [13, Eq. (5)], and define the stimulation input $\tilde{u}(t) := I_e \text{Stim}(t)$.

$$\begin{aligned}\tau_S \dot{\text{STN}}(t) &= F_S(-w_{GS} \text{GP}(t - \tau_{GS}) \\ &\quad + w_{CS} \text{Ctx}) - \text{STN}(t) + \tilde{u}(t), \\ \tau_G \dot{\text{GP}}(t) &= F_G(w_{SG} \text{STN}(t - \tau_{SG}) \\ &\quad - w_{GG} \text{GP}(t - \tau_{GG}) \\ &\quad - w_{XG} \text{Str} - \text{GP}(t)).\end{aligned}\quad (11)$$

Let STN and GP denote the physical firing rates of the SubThalamic Nucleus (STN) and Globus Pallidus pars Externa (GPe). The activation functions are

$$F_i(q) = \frac{M_i}{1 + \exp(-4q/M_i)(M_i - B_i)/B_i)}, \quad i \in \{S, G\}, \quad (12)$$

so $F_i(0) = B_i$. Define the sigmoid midpoint and its centered form as

$$\begin{aligned}c_i &:= \frac{M_i}{4} \ln\left(\frac{M_i - B_i}{B_i}\right), \\ \sigma_i(q) &:= \frac{M_i}{1 + \exp(-4q/M_i)} - \frac{M_i}{2} = \frac{M_i}{2} \tanh\left(\frac{2q}{M_i}\right).\end{aligned}\quad (13)$$

Then $F_i(q) = \sigma_i(q - c_i) + M_i/2$. Define the centered states

$$\tilde{x}_S := \text{STN} - \frac{M_S}{2}, \quad \tilde{x}_G := \text{GP} - \frac{M_G}{2},$$

and offsets

$$\begin{aligned}b_S &:= w_{CS} \text{Ctx} - \frac{w_{GS} M_G}{2} - c_S, \\ b_G &:= \frac{w_{SG} M_S}{2} - \frac{w_{GG} M_G}{2} - w_{XG} \text{Str} - c_G.\end{aligned}$$

This yields a model in which the nonlinearities σ_S and σ_G are centered and satisfy $\sigma_i(0) = 0$:

$$\begin{aligned}\tilde{z}_S(t) &:= -w_{GS} \tilde{x}_G(t - \tau_{GS}) + b_S, \\ \tilde{z}_G(t) &:= w_{SG} \tilde{x}_S(t - \tau_{SG}) - w_{GG} \tilde{x}_G(t - \tau_{GG}) + b_G, \\ \tau_S \dot{\tilde{x}}_S(t) &= \sigma_S(\tilde{z}_S(t)) - \tilde{x}_S(t) + \tilde{u}(t), \\ \tau_G \dot{\tilde{x}}_G(t) &= \sigma_G(\tilde{z}_G(t)) - \tilde{x}_G(t).\end{aligned}\quad (14)$$

Here τ_{ij} and w_{ij} denote the transmission delay and synaptic weight, respectively, from population i to population j . τ_S and τ_G denote the time constants for the STN and GPe populations, respectively. To formulate the control objective around a desired operating point, we introduce the equilibrium $(\tilde{x}_S^*, \tilde{x}_G^*)$ satisfying

$$\begin{aligned}\tilde{z}_S^* &:= -w_{GS} \tilde{x}_G^* + b_S, \\ \tilde{z}_G^* &:= w_{SG} \tilde{x}_S^* - w_{GG} \tilde{x}_G^* + b_G, \\ \tilde{x}_S^* &= \sigma_S(\tilde{z}_S^*) + \tilde{u}^*, \\ \tilde{x}_G^* &= \sigma_G(\tilde{z}_G^*).\end{aligned}$$

For $i \in \{S, G\}$, introduce the deviation variables

$$x_i(t) := \tilde{x}_i(t) - \tilde{x}_i^*, \quad z_i(t) := \tilde{z}_i(t) - \tilde{z}_i^*, \quad u(t) := \tilde{u}(t) - \tilde{u}^*,$$

and define the shifted nonlinearities

$$\delta_i(v) := \sigma_i(\tilde{z}_i^* + v) - \sigma_i(\tilde{z}_i^*).$$

The resulting system is a nonlinear DDE with the equilibrium shifted to the origin. Following [14], we introduce the delayed states $\phi_{ij}(t, s) := x_i(t - \tau_{ij}s)$ for $(i, j) \in \mathcal{I}_D$ and $s \in [0, 1]$, where $\mathcal{I}_D := \{(G, S), (S, G), (G, G)\}$. This yields the equivalent ODE-PDE representation,

$$\begin{aligned}z_S(t) &:= -w_{GS} \phi_{GS}(t, 1), \\ z_G(t) &:= w_{SG} \phi_{SG}(t, 1) - w_{GG} \phi_{GG}(t, 1), \\ \tau_S \dot{x}_S(t) &= \delta_S(z_S(t)) - x_S(t) + u(t), \\ \tau_G \dot{x}_G(t) &= \delta_G(z_G(t)) - x_G(t), \\ \dot{\phi}_{ij}(t, s) &= -\frac{1}{\tau_{ij}} \phi_{ij,s}(t, s), \\ \phi_{ij}(t, 0) &= x_i(t).\end{aligned}\quad (i, j) \in \mathcal{I}_D \quad (15)$$

Thus, the equilibrium shift only affects the nonlinear operators, while leaving the linear delayed dynamics unchanged. We now turn to robustness and performance analysis.

C. Robust Stability Analysis

We first seek the largest sector or slope bound β for which a stimulation controller can certify robust stability of (15). Thus each feasibility test includes controller synthesis: the controller is a design variable throughout the search in β . At this stage the objective is robust stabilization, with the performance channels omitted. Write

$$z_\Delta := \begin{bmatrix} z_S \\ z_G \end{bmatrix}, \quad w_\Delta := \begin{bmatrix} \delta_S(z_S) \\ \delta_G(z_G) \end{bmatrix}.$$

The shifted nonlinearities are memoryless, time invariant, monotone, and vanish at the origin. We use a common upper bound $\beta > 0$ for both channels, first retaining only the sector information

$$0 \leq v \delta_i(v) \leq \beta v^2, \quad i \in \{S, G\}, \quad (16)$$

and then also using the slope restriction

$$0 \leq \frac{\delta_i(v_1) - \delta_i(v_2)}{v_1 - v_2} \leq \beta, \quad v_1 \neq v_2. \quad (17)$$

The slope restriction implies the sector bound by taking $v_2 = 0$. We initially test these uncertainty classes on the whole real line; below we identify the trajectories of the actual sigmoids to which a given β applies. Here β denotes a sector or slope bound, independently of the beta frequency band in the example title.

1) *Sector uncertainty:* 0.492 For $D := \text{diag}(d_S, d_G) \succ 0$, the sector IQC acting on $[z_\Delta^\top \ w_\Delta^\top]^\top$ has multiplier

$$\Pi_{\text{sec}} := \begin{bmatrix} 0 & \beta D \\ \beta D & -2D \end{bmatrix}. \quad (18)$$

Its quadratic form is nonnegative pointwise by (16). Applying Lemma 1 channelwise with $\alpha = 0$ directly gives the hard dual IQC

$$\underline{\Psi}_\Delta = I_4, \quad \underline{\mathcal{V}}_\Delta = \begin{bmatrix} 0 & \beta D \\ \beta D & -2D \end{bmatrix}.$$

Thus the sector synthesis uses this hard dual IQC directly, without an ϵ completion or the ARE-based factorization. For each trial β and an admissible choice of D , we seek a controller and storage operator satisfying the stability-only synthesis LPI with this IQC. Searching in β gives the largest sector bound certified by this design procedure, denoted β_{sec}^* , together with a stabilizing controller at each feasible bound.

2) *Zames–Falb multiplier*: Zames–Falb multipliers exploit monotonicity and slope restrictions on nonlinearities [40]. To exploit (17), 0.562 replace the static weight D by the diagonal Zames–Falb (ZF) multiplier, here we use the construction provided in [31],

$$\begin{aligned} M(\zeta) &:= I - \text{diag}(H_S(\zeta), H_G(\zeta)), \\ H_i(\zeta) &:= \sum_{k=1}^3 \frac{\kappa_{ik}}{\zeta + a_{ik}}, \\ a_{ik} &> 0, \quad \kappa_{ik} \geq 0, \quad \sum_{k=1}^3 \frac{\kappa_{ik}}{a_{ik}} \leq 1. \end{aligned} \quad (19)$$

Each term has a nonnegative exponential impulse response, with integral weight κ_{ik}/a_{ik} . Thus $\|H_i\|_{L_1} \leq 1$, giving a scaled ZF multiplier valid for monotone, slope-restricted nonlinearities without assuming oddness [31, Thm. 4]. This matters because shifting the sigmoid to an arbitrary equilibrium generally removes its odd symmetry.

We choose the three pole rates from the population time constants and the recurrent GPe delay:

$$[a_{i1} \ a_{i2} \ a_{i3}] = c_i [\tau_G^{-1} \ \tau_S^{-1} \ \tau_{GG}^{-1}], \quad c_i > 0. \quad (20)$$

The poles $-a_{ik}$ supply memory on the GPe relaxation, STN relaxation, and GPe recurrent-delay scales. The reciprocal delay is a characteristic rate, not a pole of the delay operator. The dimensionless factors c_i adjust each multiplier's time scale relative to these physical scales; we initially take $c_S = c_G = 1$.

To relate the residues to the synaptic coupling strengths, define

$$\begin{aligned} [h_1 \ h_2 \ h_3] &:= \frac{\rho [|w_{GS}| \ |w_{SG}| \ |w_{GG}|]}{|w_{GS}| + |w_{SG}| + |w_{GG}|}, \\ \kappa_{ik} &:= a_{ik} h_k, \quad 0 < \rho < 1. \end{aligned}$$

This assigns the GPe relaxation term a weight proportional to the GPe-to-STN coupling, the STN relaxation term a weight proportional to the STN-to-GPe coupling, and the recurrent-delay term a weight proportional to the GPe self-coupling. The weights are nonnegative, and normalization preserves the ZF norm condition as the synaptic parameters change:

$$\sum_{k=1}^3 \frac{\kappa_{ik}}{a_{ik}} = \sum_{k=1}^3 h_k = \rho < 1, \quad \|H_i\|_{L_1} = \rho.$$

We take $\rho = 13/15$ to retain the prescribed kernel norm. This parameterization is a physically motivated choice of multiplier shape; it does not assert that the synaptic strengths determine the optimal IQC weights or guarantee an improved certificate. The factors c_i and ρ can be compared through an outer synthesis search. For fixed system parameters and these

choices, the poles and residues remain fixed during the search in β . The resulting slope IQC is

$$\Pi_{\text{ZF}}(j\omega) := \begin{bmatrix} 0 & \beta M(j\omega)^* \\ \beta M(j\omega) & -(M(j\omega) + M(j\omega)^*) \end{bmatrix}. \quad (21)$$

With all $\kappa_{ik} = 0$, this reduces to the sector multiplier. We repeat the search in β , synthesizing a controller for each trial bound using the ZF family, and denote the largest certified bound by β_{ZF}^* . Retaining the sector solution as a candidate gives $\beta_{\text{ZF}}^* \geq \beta_{\text{sec}}^*$ when the controller and storage parameterizations include those of the sector design. These bounds describe what the chosen synthesis procedure certifies; failure of a synthesis LPI does not establish that robust stabilization is impossible.

For the ZF multiplier, we use a J -spectral factorization to obtain the required hard IQC and dual filter. Since the sector bound gives $\|w_\Delta\| \leq \beta \|z_\Delta\|$, we use the PN completion as discussed in [23]

$$\Pi_\Delta := \Pi_{\text{ZF}} + \epsilon \begin{bmatrix} \beta^2 I_2 & 0 \\ 0 & -I_2 \end{bmatrix}, \quad \epsilon > 0, \quad (22)$$

The added term is a valid norm IQC. Since $M(j\omega) + M(j\omega)^* \succeq 0$, Π_Δ is strict PN and Theorem 5 supplies

$$\Pi_\Delta = \Theta_\Delta^\sim J_{2,2} \Theta_\Delta, \quad \Theta_\Delta, \Theta_\Delta^{-1} \in \mathbb{RH}_\infty.$$

For each trial β in the ZF search, select admissible multiplier weights and compute this factorization. With Θ_Δ fixed, the dual synthesis LPI in Corollary 1, with performance channels omitted, searches for a controller and storage operator certifying closed-loop robust stability. Multiplier choices can be compared outside this LPI; joint optimization of the factorization and controller is not assumed to be convex. The resulting feasible values of β therefore belong to controller–certificate pairs.

3) *Admissible trajectories*: For a controller certified at a given β , the admissible sigmoid inputs follow from (13):

$$\delta'_i(v) = \text{sech}^2(2(\tilde{z}_i^* + v)/M_i).$$

Consequently the sector and slope bounds hold globally for $\beta \geq 1$. For $0 < \beta < 1$, inversion of this derivative gives

$$\delta'_i(v) \leq \beta \iff |\tilde{z}_i^* + v| \geq \frac{M_i}{2} \text{arcosh}(\beta^{-1/2}).$$

To obtain an interval containing the equilibrium, remain on its side of the sigmoid midpoint. A simple sufficient trajectory bound is

$$|z_i(t)| < |\tilde{z}_i^*| - \frac{M_i}{2} \text{arcosh}(\beta^{-1/2}), \quad (23)$$

for $i \in \{S, G\}$ and every $t \geq 0$. The right-hand side is positive when $\beta > \delta'_i(0)$. Throughout this interval the derivative is bounded by β , so both (16) and (17) hold. For an equilibrium below the midpoint, the less restrictive one-sided condition is

$$z_i(t) \leq -\tilde{z}_i^* - \frac{M_i}{2} \text{arcosh}(\beta^{-1/2}).$$

Above the midpoint, the corresponding condition is

$$z_i(t) \geq -\tilde{z}_i^* + \frac{M_i}{2} \text{arcosh}(\beta^{-1/2}).$$

These bounds constrain the delayed firing-rate combinations in (15). A larger feasible β permits a wider interval around the same equilibrium. The local IQC certificate applies via a globally slope-restricted extension agreeing with the sigmoid on this interval, and transfers to the original system for trajectories that remain there; invariance requires a separate check.

D. Robust Performance Analysis

Having established the robustness limit through controller synthesis, fix a feasible $\beta < \beta_{ZF}^*$ and synthesize controllers with performance objectives at this smaller bound. This leaves margin relative to the robust-stability limit. For a local sigmoid certificate, also require $\beta > \max_i \delta'_i(0)$ so that (23) contains a neighborhood of the equilibrium. This choice is possible only if the two requirements leave a nonempty interval. Keep β fixed throughout the performance comparison so that all controllers are assessed on the same admissible trajectory range.

Let w_p be an additive disturbance in the stimulation channel. Using the same uncertainty channels and transport equations as above, the generalized plant is specified by

$$\begin{aligned} \tau_S \dot{x}_S &= w_{\Delta,1} - x_S + u + w_p, \\ \tau_G \dot{x}_G &= w_{\Delta,2} - x_G, \\ z_p &:= \begin{bmatrix} x_S \\ x_G \\ \mu u \end{bmatrix}, \quad \mu > 0. \end{aligned} \quad (24)$$

Thus the performance objective measures rejection of perturbations entering through the stimulation input. The effort penalty concerns the stimulation deviation $u = \tilde{u} - \tilde{u}^*$. We assess the induced L_2 gain to each signal separately, using γ_S , γ_G , and γ_u for x_S , x_G , and u , respectively. To generate controllers with different effort penalties, for zero initial deviations and controller state minimize the weighted synthesis bound γ in

$$\|x_S\|_{L_2}^2 + \|x_G\|_{L_2}^2 + \mu^2 \|u\|_{L_2}^2 \leq \gamma^2 \|w_p\|_{L_2}^2. \quad (25)$$

This directly penalizes firing-rate deviations and control effort, with the performance IQC

$$\Psi_p = I_4, \quad \mathcal{V}_p = \text{diag}(I_3, -\gamma^2). \quad (26)$$

For $\gamma > 0$, its $J_{3,1}$ -spectral factor $\text{diag}(I_3, \gamma)$ is stably invertible and can be combined with Θ_Δ in the dual synthesis condition.

For each μ , fix an admissible uncertainty multiplier at the chosen β , compute its factorization, and use Corollary 1 to synthesize the stimulation controller while minimizing γ . Keeping the multiplier fixed in each synthesis problem avoids a joint search over its factorization and the controller. Increasing μ gives more weight to reducing stimulation effort; decreasing it gives more weight to suppressing x_S and x_G . We will vary μ to obtain a family of controllers and then reanalyze each fixed controller with three separate performance outputs to certify

$$\begin{aligned} \|x_S\|_{L_2} &\leq \gamma_S \|w_p\|_{L_2}, \\ \|x_G\|_{L_2} &\leq \gamma_G \|w_p\|_{L_2}, \\ \|u\|_{L_2} &\leq \gamma_u \|w_p\|_{L_2}. \end{aligned}$$

Each gain bound is minimized in its own analysis problem, for the same controller, disturbance channel, and uncertainty bound β . We will report all three bounds and compare γ_u with γ_S and γ_G separately, showing how the effort–attenuation tradeoff differs between the two populations. The weighted γ serves only as the synthesis objective; the reported gains refer to the individual, unweighted signals. For the original sigmoid with $\beta < 1$, each comparison is restricted to trajectories satisfying (23); the L_2 -gain bounds alone do not establish this pointwise condition.

IX. CONCLUSION

REFERENCES

- [1] Joaquin Carrasco and Peter Seiler. Conditions for the equivalence between IQC and graph separation stability results. *International Journal of Control*, 2018. Published online in 2018.
- [2] Ruth F. Curtain and Keith Glover. Robust stabilization of infinite dimensional systems by finite dimensional controllers. *Systems & Control Letters*, 7(1):41–47, 1986.
- [3] Ruth F. Curtain and Anthony J. Pritchard, editors. *Infinite Dimensional Linear Systems Theory*, volume 8 of *Lecture Notes in Control and Information Sciences*. Springer-Verlag, Berlin, Heidelberg, 1978.
- [4] Ruth F. Curtain and Hans Zwart. *An Introduction to Infinite-Dimensional Linear Systems Theory*, volume 21 of *Texts in Applied Mathematics*. Springer-Verlag, New York, 1995.
- [5] A. Das, S. Shivakumar, M. Peet, and S. Weiland. Robust Analysis of Uncertain ODE-PDE Systems Using PI Multipliers, PIEs and LPIs. In *2020 59th IEEE Conference on Decision and Control (CDC)*, pages 634–639, 2020.
- [6] Emilia Fridman and Anatoly Blighovsky. Robust sampled-data control of a class of semilinear parabolic systems. *Automatica*, 48(5):826–836, 2012.
- [7] Bin Hu, Márcio J. Lacerda, and Peter Seiler. Robustness analysis of uncertain discrete-time systems with dissipation inequalities and integral quadratic constraints. *International Journal of Robust and Nonlinear Control*, 27(11):1940–1962, 2017.
- [8] I. Köse and C. Scherer. Robust L_2 -gain feedforward control of uncertain systems using dynamic IQCs. *International Journal of Robust and Nonlinear Control*, 19(11):1224–1247, 2009.
- [9] Miroslav Krstić and Andrey Smyshlyaev. *Boundary Control of PDEs: A Course on Backstepping Designs*, volume 16 of *Advances in Design and Control*. Society for Industrial and Applied Mathematics, 2008.
- [10] Thijs Lenssen, Aleksandr Talitskii, Matthew Peet, and Amritam Das. A μ -Analysis and Synthesis Framework for Partial Integral Equations using IQCs, 2025.
- [11] A. Megretski and A. Rantzer. System analysis via integral quadratic constraints. *IEEE Transactions on Automatic Control*, 42(6):819–830, 1997.
- [12] Gjerrit Meinsma. J-spectral factorization and equalizing vectors. *Systems & Control Letters*, 25(4):243–249, 1995.
- [13] Alejo J. Nevado Holgado, John R. Terry, and Rafal Bogacz. Conditions for the Generation of Beta Oscillations in the Subthalamic Nucleus–Globus Pallidus Network. *The Journal of Neuroscience*, 30(37):12340–12352, 2010.
- [14] M. Peet. Representation of Networks and Systems with Delay DDEs, DDFs, ODE-PDEs and PIEs, 2020.
- [15] Matthew M. Peet. A partial integral equation (PIE) representation of coupled linear PDEs and scalable stability analysis using LMIs. *Automatica*, 125:109473, 2021.
- [16] Madalina Petcu and Roger Temam. Control for the Sine-Gordon Equation. *ESAIM: Control, Optimisation and Calculus of Variations*, 10(4):553–573, 2004.
- [17] Harald Pfifer and Peter Seiler. Robustness analysis of linear parameter varying systems using integral quadratic constraints. *International Journal of Robust and Nonlinear Control*, 25(15):2843–2864, 2015.
- [18] Harald Pfifer and Peter Seiler. Less conservative robustness analysis of linear parameter varying systems using integral quadratic constraints. *International Journal of Robust and Nonlinear Control*, 26(16):3580–3594, 2016.
- [19] R. M. Redheffer. On a certain linear fractional transformation. *Journal of Mathematics and Physics*, 39(1-4):269–286, 1960.

- [20] C. Scherer. Robust controller synthesis is convex for systems without control channel uncertainties. In P. Van den Hof, C. Scherer, and P. Heuberger, editors, *Model Based Control: Bridging Rigorous Theory and Advanced Technology*. Springer, New York, 2009.
- [21] C. Scherer and I. Köse. Robustness with dynamic integral quadratic constraints: an exact state-space characterization of nominal stability with applications to robust estimation. *Automatica*, 44(7):1666–1675, 2008.
- [22] Carsten W. Scherer and Joost Veenman. Stability analysis by dynamic dissipation inequalities: On merging frequency-domain techniques with time-domain conditions. *Systems & Control Letters*, 121:7–15, 2018.
- [23] Peter Seiler. Stability analysis with dissipation inequalities and integral quadratic constraints. *IEEE Transactions on Automatic Control*, 60(6):1704–1709, 2015.
- [24] S. Shivakumar, A. Das, and M. Peet. PIETOOLS: A Matlab Toolbox for Manipulation and Optimization of Partial Integral Operators. In *2020 American Control Conference (ACC)*, pages 2667–2672, 2020.
- [25] S. Shivakumar, A. Das, S. Weiland, and M. Peet. Extension of the Partial Integral Equation Representation to GPDE Input–Output Systems. *IEEE Transactions on Automatic Control*, 70(5):3240–3255, 2025.
- [26] Sachin Shivakumar, Amritam Das, and Matthew Peet. Dual representations and H_∞ -optimal control of partial differential equations. *IEEE Transactions on Automatic Control*, 71(8):4956–4971, 2026.
- [27] Sachin Shivakumar, Declan Jagt, Danilo Braghini, Amritam Das, Yulia Peet, and Matthew Peet. PIETOOLS 2025: User manual, 2026. Version 2, revised 15 March 2026; first submitted in 2025.
- [28] Olof Staffans. *Well-Posed Linear Systems*, volume 103 of *Encyclopedia of Mathematics and its Applications*. Cambridge University Press, 2005.
- [29] A. Talitckii, M. Peet, and P. Seiler. Integral Quadratic Constraints with Infinite-Dimensional Channels. In *2023 American Control Conference (ACC)*, pages 1576–1583, 2023.
- [30] Samarth Toolhally, Joeri Roelofs, Siep Weiland, and Amritam Das. A digital twin of evaporative thermo-fluidic process in the fixation unit of dod inkjet printers. *Mechatronics*, 118:103558, 2026.
- [31] Matthew C. Turner. Zames-falb multipliers: don’t panic, 2022.
- [32] Giorgio Valmorbida, Mohamadreza Ahmadi, and Antonis Papachristodoulou. Stability analysis for a class of partial differential equations via semidefinite programming. *IEEE Transactions on Automatic Control*, 61(6):1649–1654, 2016.
- [33] Bert van Keulen. *H_∞ -Control for Distributed Parameter Systems: A State-Space Approach*. Systems & Control: Foundations & Applications. Birkhäuser, Boston, 1993.
- [34] J. Veenman, C. Scherer, and H. Köroğlu. Robust stability and performance analysis based on integral quadratic constraints. *European Journal of Control*, 31:1–32, 2016.
- [35] Joost Veenman and Carsten W. Scherer. IQC-synthesis with general dynamic multipliers. *International Journal of Robust and Nonlinear Control*, 24(17):3027–3056, 2014.
- [36] Raghu Venkataraman and Peter Seiler. Convex LPV synthesis of estimators and feedforwards using duality and integral quadratic constraints. *International Journal of Robust and Nonlinear Control*, 28(3):953–975, 2018.
- [37] Shu Wang, Harald Pfifer, and Peter Seiler. Robust synthesis for linear parameter varying systems using integral quadratic constraints. *Automatica*, 68:111–118, 2016.
- [38] Jan C. Willems. Dissipative dynamical systems—Part I: General theory. *Archive for Rational Mechanics and Analysis*, 45(5):321–351, 1972.
- [39] Chengzhi Yuan and Fen Wu. Dynamic iq-c-based control of uncertain lft systems with time-varying state delay. *IEEE Transactions on Cybernetics*, 46(12):3320–3329, 2016.
- [40] G. Zames and P. Falb. Stability conditions for systems with monotone and slope-restricted nonlinearities. *SIAM Journal on Control*, 6(1):89–109, 1968.
- [41] Kemin Zhou, John Comstock Doyle, Keith Glover, et al. *Robust and optimal control*, volume 40. Prentice hall New Jersey, 1996.

APPENDIX A PROOFS

A. Proof of Theorem 1

Suppose that (2) holds. This implies,

$$\left\langle p_T \Psi_\Delta \begin{bmatrix} P_\Delta \\ I \end{bmatrix} w_\Delta, p_T \mathcal{M}_{\mathcal{V}_\Delta} \Psi_\Delta \begin{bmatrix} P_\Delta \\ I \end{bmatrix} w_\Delta \right\rangle_{\mathbf{L}_{[a,b]}} + \left\langle p_T \Psi_P \begin{bmatrix} P_P \\ 0 \end{bmatrix} w_\Delta, p_T \mathcal{M}_{\mathcal{V}_P} \Psi_P \begin{bmatrix} P_P \\ 0 \end{bmatrix} w_\Delta \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon \|p_T w_\Delta\|_{\mathbf{L}_{[a,b]}}^2.$$

By Definition 11 we can then infer,

$$\left\langle p_T \Psi_\Delta \begin{bmatrix} P_\Delta w_\Delta \\ w_\Delta \end{bmatrix}, p_T \mathcal{M}_{\mathcal{V}_\Delta} \Psi_\Delta \begin{bmatrix} P_\Delta w_\Delta \\ w_\Delta \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon \|p_T w_\Delta\|_{\mathbf{L}_{[a,b]}}^2.$$

Then, using Theorem 6 in [29], we find that **robust stability** is guaranteed. Moreover, rewriting (2) as,

$$\left\langle p_T \Psi_\Delta \begin{bmatrix} z_\Delta \\ w_\Delta \end{bmatrix}, p_T \mathcal{M}_{\mathcal{V}_\Delta} \Psi_\Delta \begin{bmatrix} z_\Delta \\ w_\Delta \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} + \left\langle p_T \Psi_P \begin{bmatrix} z_P \\ w_P \end{bmatrix}, p_T \mathcal{M}_{\mathcal{V}_P} \Psi_P \begin{bmatrix} z_P \\ w_P \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon (\|p_T w_\Delta\|_{\mathbf{L}_{[a,b]}}^2 + \|p_T w_P\|_{\mathbf{L}_{[a,b]}}^2),$$

and given that Ψ_Δ and $\mathcal{M}_{\mathcal{V}_\Delta}$ satisfy the IQC in Definition 10 leads to,

$$\left\langle p_T \Psi_P \begin{bmatrix} z_P \\ w_P \end{bmatrix}, p_T \mathcal{M}_{\mathcal{V}_P} \Psi_P \begin{bmatrix} z_P \\ w_P \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon \|p_T w_P\|_{\mathbf{L}_{[a,b]}}^2.$$

Thus by Definition 11 **robust performance** is achieved.

B. Proof of Theorem 3

1) $\implies$ 2) Assume there exists a $\rho < 1$ such that $\|p_T G_0 \tilde{w}\|_{\mathbf{L}} \leq \rho \|p_T \tilde{w}\|_{\mathbf{L}}$ for all $\tilde{w} \in \mathbf{L}$ and $T \geq 0$. We need to show that $\mathcal{I} \leq -\epsilon \|p_T w\|_{\mathbf{L}}^2$ for some $\epsilon > 0$. First since Γ is causally boundedly invertible, $\Theta \begin{bmatrix} P \\ I \end{bmatrix} w = \begin{bmatrix} G_0 \\ I \end{bmatrix} \Gamma w$, and hence $\mathcal{I} = \|p_T G_0 \Gamma w\|_{\mathbf{L}}^2 - \|p_T \Gamma w\|_{\mathbf{L}}^2$. By statement 1 and causality of G_0 ,

$$\mathcal{I} \leq -(1 - \rho^2) \|p_T \Gamma w\|_{\mathbf{L}}^2 \leq -\frac{1 - \rho^2}{\|\Gamma^{-1}\|_{\mathcal{L}(\mathbf{L})}^2} \|p_T w\|_{\mathbf{L}}^2,$$

where the second inequality follows from $\|p_T w\|_{\mathbf{L}} = \|p_T \Gamma^{-1} p_T \Gamma w\|_{\mathbf{L}} \leq \|\Gamma^{-1}\|_{\mathcal{L}(\mathbf{L})} \|p_T \Gamma w\|_{\mathbf{L}}$. Thus, $\epsilon = \frac{(1 - \rho^2)}{\|\Gamma^{-1}\|_{\mathcal{L}(\mathbf{L})}^2} > 0$.

2) $\implies$ 1) Assume that there exists a $\epsilon > 0$ such that $\mathcal{I} \leq -\epsilon \|p_T w\|_{\mathbf{L}}^2$ for every w , and every $T \geq 0$. We need to establish there exists a $\rho < 1$ such that $\|p_T G_0 \tilde{w}\|_{\mathbf{L}} \leq \rho \|p_T \tilde{w}\|_{\mathbf{L}}$ for all $\tilde{w} \in \mathbf{L}$ and $T \geq 0$. For arbitrary $\tilde{w}$, set $w = \Gamma^{-1} \tilde{w}$. Then

$$\|p_T G_0 \tilde{w}\|_{\mathbf{L}}^2 - \|p_T \tilde{w}\|_{\mathbf{L}}^2 \leq -\epsilon \|p_T \Gamma^{-1} \tilde{w}\|_{\mathbf{L}}^2.$$

Since Γ is bounded and causal, $\|p_T \tilde{w}\|_{\mathbf{L}} \leq \|\Gamma\|_{\mathcal{L}(\mathbf{L})} \|p_T \Gamma^{-1} \tilde{w}\|_{\mathbf{L}}$. Therefore,

$$\|p_T G_0 \tilde{w}\|_{\mathbf{L}}^2 \leq \left(1 - \frac{\epsilon}{\|\Gamma\|_{\mathcal{L}(\mathbf{L})}^2}\right) \|p_T \tilde{w}\|_{\mathbf{L}}^2.$$

Choose $\rho < 1$ such that $1 - \epsilon/\|\Gamma\|^2 \leq \rho^2 < 1$, then $\|p_T G_0 \tilde{w}\|_{\mathbf{L}} \leq \rho \|p_T \tilde{w}\|_{\mathbf{L}}$ for every $\tilde{w}$ and $T \geq 0$.

1) $\iff$ 3) We show that there exists a causally boundedly invertible $\underline{\Gamma}$ such that

$$\mathbf{D}(\Theta) \begin{bmatrix} P^\top \\ I \end{bmatrix} \underline{w} = \begin{bmatrix} G_0^\top \\ I \end{bmatrix} \underline{\Gamma} w.$$

then we can apply the previous arguments to show 1) $\iff$ 3). Define

$$S := \Theta_{11} - \Theta_{12} \Theta_{22}^{-1} \Theta_{21}.$$

Since Θ and Θ_{22} are boundedly invertible, so is S by Schur complement. Block inversion of $\mathbf{D}(\Theta)$ gives

$$\mathbf{D}(\Theta)_{21} = (S^\top)^{-1} \Theta_{21}^\top (\Theta_{22}^\top)^{-1}, \quad \mathbf{D}(\Theta)_{22} = (S^\top)^{-1}.$$

Therefore, define

$$\underline{\Gamma} := \mathbf{D}(\Theta)_{21} P^\top + \mathbf{D}(\Theta)_{22} = (S^\top)^{-1} (I + P \Theta_{22}^{-1} \Theta_{21})^\top.$$

Now $\Gamma = \Theta_{22} (I + \Theta_{22}^{-1} \Theta_{21} P)$. Thus, invertibility of Γ and the push-through rule imply that $I + P \Theta_{22}^{-1} \Theta_{21}$, and hence $\underline{\Gamma}$, is causally boundedly invertible. Normalizing the upper block gives

$$\mathbf{D}(\Theta) \begin{bmatrix} P^\top \\ I \end{bmatrix} \underline{w} = \begin{bmatrix} G_0^\top \\ I \end{bmatrix} \underline{\Gamma} w, \quad \underline{G}_0 := (\Theta_{22}^\top)^{-1} (P^\top \underline{\Gamma}^{-1} + \Theta_{12}^\top).$$

Using

$$\underline{\Gamma}^{-\top} = S(I + P\Theta_{22}^{-1}\Theta_{21})^{-1}, \quad (I + P\Theta_{22}^{-1}\Theta_{21})^{-1}P\Theta_{22}^{-1} = P\Gamma^{-1},$$

where the second identity is the push-through rule, yields

$$\begin{aligned} \underline{G}_0 &= (\Theta_{22}^\top)^{-1}(P^\top \underline{\Gamma}^{-1} + \Theta_{12}^\top) \\ &= \left(\underline{\Gamma}^{-\top} P \Theta_{22}^{-1} + \Theta_{12} \Theta_{22}^{-1} \right)^\top \\ &= \left[S(I + P\Theta_{22}^{-1}\Theta_{21})^{-1} P \Theta_{22}^{-1} + \Theta_{12} \Theta_{22}^{-1} \right]^\top \\ &= \left[S P \Gamma^{-1} + \Theta_{12} \Theta_{22}^{-1} \right]^\top \\ &= \left[(S P + \Theta_{12} \Theta_{22}^{-1} \Gamma) \Gamma^{-1} \right]^\top. \end{aligned}$$

We now show that the first part reduces to

$$\begin{aligned} S P + \Theta_{12} \Theta_{22}^{-1} \Gamma &= \Theta_{11} P - \Theta_{12} \Theta_{22}^{-1} \Theta_{21} P \\ &\quad + \Theta_{12} \Theta_{22}^{-1} \Theta_{21} P + \Theta_{12} \\ &= \Theta_{11} P + \Theta_{12}. \end{aligned}$$

Consequently,

$$\underline{G}_0 = \left[(\Theta_{11} P + \Theta_{12}) \Gamma^{-1} \right]^\top = G_0^\top,$$

and hence

$$\mathbf{D}(\Theta) \begin{bmatrix} P^\top \\ I \end{bmatrix} \underline{w} = \begin{bmatrix} G_0^\top \\ I \end{bmatrix} \Gamma w.$$

By Theorem 2, G_0 and $G_0^\top$ have the same induced L_2 -gain. Hence, for any ρ ,

$$\|p_T G_0 \tilde{w}\|_{\mathbf{L}} \leq \rho \|p_T \tilde{w}\|_{\mathbf{L}} \quad \text{for all } \tilde{w} \text{ and } T \geq 0 \quad \iff \quad \|p_T G_0^\top \tilde{w}\|_{\mathbf{L}} \leq \rho \|p_T \tilde{w}\|_{\mathbf{L}} \quad \text{for all } \tilde{w} \text{ and } T \geq 0.$$

Since $\underline{\Gamma}$ is causally boundedly invertible, the same arguments used for 1) $\implies$ 2) and 2) $\implies$ 1), with G_0 , Γ , w , and $\mathcal{I}$ replaced by $G_0^\top$, $\underline{\Gamma}$, $\underline{w}$, and $\underline{\mathcal{I}}$, respectively, establish 1) $\iff$ 3).

a) *Realization of G_0 .* Let $\mathbf{x}$ collect the plant, filter, and any controller states, so that the controller is already included in the primal operators. Partition the filtered realization as

$$\begin{bmatrix} \mathcal{T} \dot{\mathbf{x}} \\ \tilde{z} \\ \tilde{w} \end{bmatrix} = \begin{bmatrix} \mathcal{A} & \mathcal{B} \\ \mathcal{C}_z & \mathcal{D}_z \\ \mathcal{C}_w & \mathcal{D}_w \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ w \end{bmatrix}.$$

If $\mathcal{D}_w$ is invertible, eliminating w gives the realization

$$G_0 \sim \{ \mathcal{T}, \mathcal{A} - \mathcal{B} \mathcal{D}_w^{-1} \mathcal{C}_w, \mathcal{B} \mathcal{D}_w^{-1}, \mathcal{C}_z - \mathcal{D}_z \mathcal{D}_w^{-1} \mathcal{C}_w, \mathcal{D}_z \mathcal{D}_w^{-1} \}, \quad (27)$$

with input $\tilde{w}$ and output $\tilde{z}$.

C. Proof of Corollary 4

The last row of the primal component block gives the PI coordinate map

$$\mathcal{M}_\Gamma := \begin{bmatrix} I & 0 & 0 \\ 0 & I & 0 \\ \mathcal{D}_{21} \mathcal{C}_P & \mathcal{C}_2 & \mathcal{D}_{21} \mathcal{D}_P + \mathcal{D}_{22} \end{bmatrix}, \quad \mathcal{M}_{\Gamma^{-1}} := \begin{bmatrix} I & 0 & 0 \\ 0 & I & 0 \\ -\Lambda \mathcal{D}_{21} \mathcal{C}_P & -\Lambda \mathcal{C}_2 & \Lambda \end{bmatrix}.$$

Both maps belong to Π_4 , and direct block multiplication verifies that they are inverses. In particular,

$$\begin{bmatrix} \mathbf{x}_P \\ \mathbf{x}_\Theta \\ \tilde{w} \end{bmatrix} = \mathcal{M}_\Gamma \begin{bmatrix} \mathbf{x}_P \\ \mathbf{x}_\Theta \\ w \end{bmatrix}, \quad \begin{bmatrix} \mathbf{x}_P \\ \mathbf{x}_\Theta \\ w \end{bmatrix} = \mathcal{M}_{\Gamma^{-1}} \begin{bmatrix} \mathbf{x}_P \\ \mathbf{x}_\Theta \\ \tilde{w} \end{bmatrix}.$$

Thus $\mathcal{M}_\Gamma$ is the PI congruence on the augmented state–input coordinates; the causal map $\Gamma : w \mapsto \tilde{w}$ in Theorem 3 is its induced input–output map. The construction in the proof of Theorem 3 shows that the corresponding coordinate map from the dual component block is then invertible as well.

The primal and dual LPIs are the KYP inequalities for their respective component blocks. Integration with the nonnegative storage gives statements 2 and 3 of Theorem 3, respectively. Since these are equivalent, either LPI yields the strict primal hard IQC (2). Theorem 1 then gives robust stability and robust performance.

D. Proof of Corollary 1

Since $\mathcal{P} \succ 0$, define $\mathcal{K}_P$ and $\mathcal{K}_\Theta$ by

$$\mathcal{Z}^* = [\mathcal{K}_P \quad \mathcal{K}_\Theta] \mathcal{P}.$$

The feedback-dependent rows of the dual component block give

$$\underline{\mathcal{A}} := \underline{\mathcal{A}}_0 + \begin{bmatrix} \mathcal{K}_P^* \\ \mathcal{K}_\Theta^* \end{bmatrix} \begin{bmatrix} \mathcal{B}_u^* & 0 \end{bmatrix}, \quad \underline{\mathcal{B}} := \underline{\mathcal{B}}_0 + \begin{bmatrix} \mathcal{K}_P^* \\ \mathcal{K}_\Theta^* \end{bmatrix} \mathcal{D}_{zu}^*.$$

Then (7) is the dual KYP inequality for the closed-loop filtered system $\mathbf{D}(\Theta) \begin{bmatrix} (K \star P)^\top \\ I \end{bmatrix}$. From the last row of the primal component block, its PI congruence transformation and inverse are

$$\mathcal{M}_\Gamma := \begin{bmatrix} I & 0 & 0 \\ 0 & I & 0 \\ \mathcal{D}_{21}(\mathcal{C}_P + \mathcal{D}_{zu}\mathcal{K}_P) & \mathcal{C}_2 + \mathcal{D}_{21}\mathcal{D}_{zu}\mathcal{K}_\Theta & \mathcal{D}_{21}\mathcal{D}_P + \mathcal{D}_{22} \end{bmatrix},$$

$$\mathcal{M}_{\Gamma^{-1}} := \begin{bmatrix} I & 0 & 0 \\ 0 & I & 0 \\ -\Lambda\mathcal{D}_{21}(\mathcal{C}_P + \mathcal{D}_{zu}\mathcal{K}_P) & -\Lambda(\mathcal{C}_2 + \mathcal{D}_{21}\mathcal{D}_{zu}\mathcal{K}_\Theta) & \Lambda \end{bmatrix}.$$

Equivalently, the primal filtered-input row is

$$\begin{bmatrix} \mathbf{x}_P \\ \mathbf{x}_\Theta \\ \tilde{w} \end{bmatrix} = \mathcal{M}_\Gamma \begin{bmatrix} \mathbf{x}_P \\ \mathbf{x}_\Theta \\ w \end{bmatrix}.$$

The last row of the dual component block gives the differently structured dual map. By the primal–dual invertibility equivalence in Theorem 3,

$$\underline{\Lambda} := (\underline{\mathcal{D}}_{21}\mathcal{D}_P^* + \underline{\mathcal{D}}_{22})^{-1} \in \Pi_4,$$

and

$$\mathcal{M}_{\underline{\Gamma}} := \begin{bmatrix} I & 0 & 0 \\ 0 & I & 0 \\ \underline{\mathcal{D}}_{21}\mathcal{B}_P^* & \underline{\mathcal{C}}_2 & \underline{\mathcal{D}}_{21}\mathcal{D}_P^* + \underline{\mathcal{D}}_{22} \end{bmatrix}, \quad \mathcal{M}_{\underline{\Gamma}^{-1}} := \begin{bmatrix} I & 0 & 0 \\ 0 & I & 0 \\ -\underline{\Lambda}\underline{\mathcal{D}}_{21}\mathcal{B}_P^* & -\underline{\Lambda}\underline{\mathcal{C}}_2 & \underline{\Lambda} \end{bmatrix}.$$

That row acts as

$$\begin{bmatrix} \mathbf{x}_P \\ \mathbf{x}_\Theta \\ \tilde{w} \end{bmatrix} = \mathcal{M}_{\underline{\Gamma}} \begin{bmatrix} \mathbf{x}_P \\ \mathbf{x}_\Theta \\ w \end{bmatrix}.$$

Unlike $\mathcal{M}_\Gamma$, the dual map has no explicit controller-gain blocks; the gains enter the dual state and input operators above. The analysis map is recovered by setting $\mathcal{K}_P = \mathcal{K}_\Theta = 0$. These PI matrices act on the augmented state–input coordinates and induce the causal maps Γ and $\underline{\Gamma}$ used in Theorem 3; they are not those input–output maps themselves. All four coordinate maps lie in Π_4 . Hence the dual KYP inequality gives statement 3 of Theorem 3, which is equivalent to the strict primal hard IQC in statement 2. Therefore Theorem 1 proves robust stability and robust performance of $K \star P \star \Delta$.

E. Proof of Lemma 1

Suppose that the dual IQC

$$\left\langle p_T \begin{bmatrix} P_\Delta^\top w_\Delta \\ w_\Delta \end{bmatrix}, p_T \mathcal{M}_{\underline{\mathcal{V}}_\Delta} \begin{bmatrix} P^\top w_\Delta \\ w_\Delta \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}}$$

is parametrized according to (9). We need to show that Δ satisfies the primal IQC. First, observe the dual IQC can be factorized as

$$\underline{\mathcal{V}}_\Delta = \frac{1}{2} \begin{bmatrix} \beta & -1 \\ -\alpha & 1 \end{bmatrix}^* \begin{bmatrix} \mathcal{S} & I \\ \mathcal{S} & -I \end{bmatrix}^* \begin{bmatrix} I & 0 \\ 0 & -I \end{bmatrix} \underbrace{\begin{bmatrix} \mathcal{S} & I \\ \mathcal{S} & -I \end{bmatrix}}_{\underline{\Theta}_1} \underbrace{\begin{bmatrix} \beta & -1 \\ -\alpha & 1 \end{bmatrix}}_{\underline{\Theta}_2} \quad (28)$$

Then, we use the multiplicative and involutive property of the dual operation such that $\mathbf{D}(\Theta_1\Theta_2) = \mathbf{D}(\Theta_1)\mathbf{D}(\Theta_2)$ for any two invertible Θ_1 and Θ_2 and $\mathbf{D}(\mathbf{D}(\Theta)) = \Theta$ to recover the primal IQC. Since $\underline{\mathcal{S}}_0 \succeq 0$, $\underline{\mathcal{S}}$ is closed under inversion with $\underline{\mathcal{S}}_0^{-1} \succeq 0$. Applying these properties to the two factors of $\underline{\Theta}$ in (28) gives

$$\begin{aligned} \Theta &= \mathbf{D}(\underline{\Theta}) = \mathbf{D}\left(\frac{1}{\sqrt{2}} \begin{bmatrix} I & \underline{\mathcal{S}} \\ I & -\underline{\mathcal{S}} \end{bmatrix}\right) \mathbf{D}\left(\begin{bmatrix} \beta & -1 \\ -\alpha & 1 \end{bmatrix}\right) \\ &= \frac{1}{\sqrt{2}(\beta - \alpha)} \begin{bmatrix} -\underline{\mathcal{S}}^{-1} & -I \\ -\underline{\mathcal{S}}^{-1} & I \end{bmatrix} \begin{bmatrix} \beta & -1 \\ -\alpha & 1 \end{bmatrix}. \end{aligned}$$

We omit $\frac{1}{(\beta-\alpha)} > 0$ therefore,

$$\mathcal{V}_\Delta = \begin{bmatrix} \beta & -1 \\ -\alpha & 1 \end{bmatrix}^* \begin{bmatrix} 0 & \underline{\mathcal{S}}^{-1} \\ \underline{\mathcal{S}}^{-1} & 0 \end{bmatrix} \begin{bmatrix} \beta & -1 \\ -\alpha & 1 \end{bmatrix}$$

For completion we use the proof in [29] Then, for all $s \in [a, b]$ and $t > 0$, we have that

$$(\beta z_\Delta(s, t) - \phi(z_\Delta(s, t)))(\phi(z_\Delta(s, t)) - \alpha z_\Delta(s, t)) \geq 0,$$

and hence

$$(\beta z_\Delta(s, t) - \phi(z_\Delta(s, t)))\underline{\mathcal{S}}_0^{-1}(s)(\phi(z_\Delta(s, t)) - \alpha z_\Delta(s, t)) \geq 0.$$

Therefore, Δ satisfies the IQC defined by $\underline{\Psi}_\Delta = I$ and (9).